

The Maximal Overlap Discrete Wavelet Transform

5.0 Introduction

In this chapter we describe a modified version of the discrete wavelet transform called the maximal overlap DWT (MODWT). The name comes from the literature on the Allan variance (see Section 8.6), where the MODWT based upon the Haar wavelet filter has been used since the 1970s (a good entry point to this body of literature is Greenhall, 1991; see also Percival and Guttorp, 1994). Transforms that are essentially the same as – or have some similarity to – the MODWT have been discussed in the wavelet literature under the names ‘undecimated DWT’ (Shensa, 1992), ‘shift invariant DWT’ (Beylkin, 1992; Lang *et al.*, 1995), ‘wavelet frames’ (Unser, 1995), ‘translation invariant DWT’ (Coifman and Donoho, 1995; Liang and Parks, 1996; Del Marco and Weiss, 1997), ‘stationary DWT’ (Nason and Silverman, 1995), ‘time invariant DWT’ (Pesquet *et al.*, 1996) and ‘non-decimated DWT’ (Bruce and Gao, 1996a). While all these names are lacking somewhat as an accurate description of the transform, we prefer ‘maximal overlap DWT’ mainly because it leads to an acronym that is easy to say (‘mod WT’) and carries the connotation of a ‘modification of the DWT.’

In contrast to the orthonormal partial DWT, the MODWT of level J_0 for a time series $\mathbf{X}$ is a highly redundant nonorthogonal transform yielding the column vectors $\tilde{\mathbf{W}}_1, \tilde{\mathbf{W}}_2, \dots, \tilde{\mathbf{W}}_{J_0}$ and $\tilde{\mathbf{V}}_{J_0}$, each of dimension N . The vector $\tilde{\mathbf{W}}_j$ contains the MODWT wavelet coefficients associated with changes in $\mathbf{X}$ on a scale of $\tau_j = 2^{j-1}$, while $\tilde{\mathbf{V}}_{J_0}$ contains the MODWT scaling coefficients associated with variations at scales $\lambda_{J_0} = 2^{J_0}$ and higher. Like the DWT, the MODWT is defined in terms of a computationally efficient pyramid algorithm. There are five important properties that distinguish the MODWT from the DWT:

- [1] While the partial DWT of level J_0 restricts the sample size to an integer multiple of 2^{J_0} , the MODWT of level J_0 is well defined for any sample size N . When N is an integer multiple of 2^{J_0} , the partial DWT can be computed using $O(N)$ multiplications, whereas the corresponding MODWT requires $O(N \log_2 N)$ multiplications. There is thus a computational price to pay for using the MODWT, but its computational burden is the same as the widely used fast Fourier transform algorithm and hence is usually quite acceptable.

- [2] As is true for the DWT, the MODWT can be used to form a multiresolution analysis (MRA). In contrast to the usual DWT, the details $\tilde{D}_j$ and smooth $\tilde{S}_{J_0}$ of this MRA are such that circularly shifting the time series by any amount will circularly shift each detail and smooth by a corresponding amount.
- [3] In contrast to the DWT, the MODWT details and smooths are associated with zero phase filters, thus making it easy to line up features in an MRA with the original time series meaningfully.
- [4] As is true for the DWT, the MODWT can be used to form an analysis of variance (ANOVA) based upon the wavelet and scaling coefficients; in contrast to the DWT, the MODWT details and smooths cannot be used to form such an analysis.
- [5] Whereas a time series and a circular shift of the series can have different DWT-based empirical power spectra (as demonstrated by Equations (63a) and (63b)), the corresponding MODWT-based spectra are the same. In fact, we can obtain the MODWT of a circularly shifted time series by just applying a similar shift to each of the components $\tilde{W}_j$ and $\tilde{V}_{J_0}$ of the MODWT of the original series; by contrast, the DWT of a circular series cannot in general be obtained by any circular shift of the components W_j and V_{J_0} of the DWT of the original series.

After we illustrate in Section 5.1 the problems that can arise with the ordinary DWT when we circularly shift $\mathbf{X}$, we define the basic MODWT wavelet and scaling filters in Section 5.2 and use these to motivate and prove the basic results for the MODWT in Sections 5.3 and 5.4. We present a pyramid algorithm for efficiently computing the MODWT in Section 5.5, followed by a simple example in Section 5.6 of how the MODWT treats a circularly shifted time series. We then illustrate the use of the MODWT through several examples in Sections 5.7 to 5.10. We close with a discussion of some practical considerations (Section 5.11) and a summary of the key results in this chapter (Section 5.12).

5.1 Effect of Circular Shifts on the DWT

Our interest in the MODWT stems from the fact that an analysis of a time series using the DWT can depend critically on where we ‘break into’ the series, i.e., what we take as a starting point or origin for analysis. We have noted the sensitivity of the DWT to such effects briefly in the context of the discrete Haar wavelet empirical power spectrum (see Equations (63a) and (63b)), but here we wish to show how the starting point can affect the DWT and its associated MRA (for comparison, we will revisit the example presented below in Section 5.6 using the MODWT).

The bottom row of Figure 161 has duplicate plots of both a time series $\mathbf{X}$ (first and third columns) and a version of the series circularly shifted by five units, i.e., $T^5\mathbf{X}$ (second and fourth columns). The time series has a sample size of $N = 128$ and consists of a ‘bump’ extending over six values surrounded by zeros:

$$X_t = \begin{cases} \frac{1}{2} \cos(\frac{3\pi t}{16} + 0.08), & t = 40, \dots, 45; \\ 0, & t = 0, \dots, 39 \text{ and } 46, \dots, 127. \end{cases}$$

The top five rows of Figure 161 show partial DWTs of order $J_0 = 4$ for $\mathbf{X}$ and $T^5\mathbf{X}$ (first two columns) along with the related MRAs (last two columns). The DWTs are all based upon the LA(8) wavelet filter. The DWT coefficients W_j and V_4 are plotted as described in Section 4.8 so that they are aligned with the original time series based upon the phase properties of the LA(8) filter. Note that, because these coefficients are

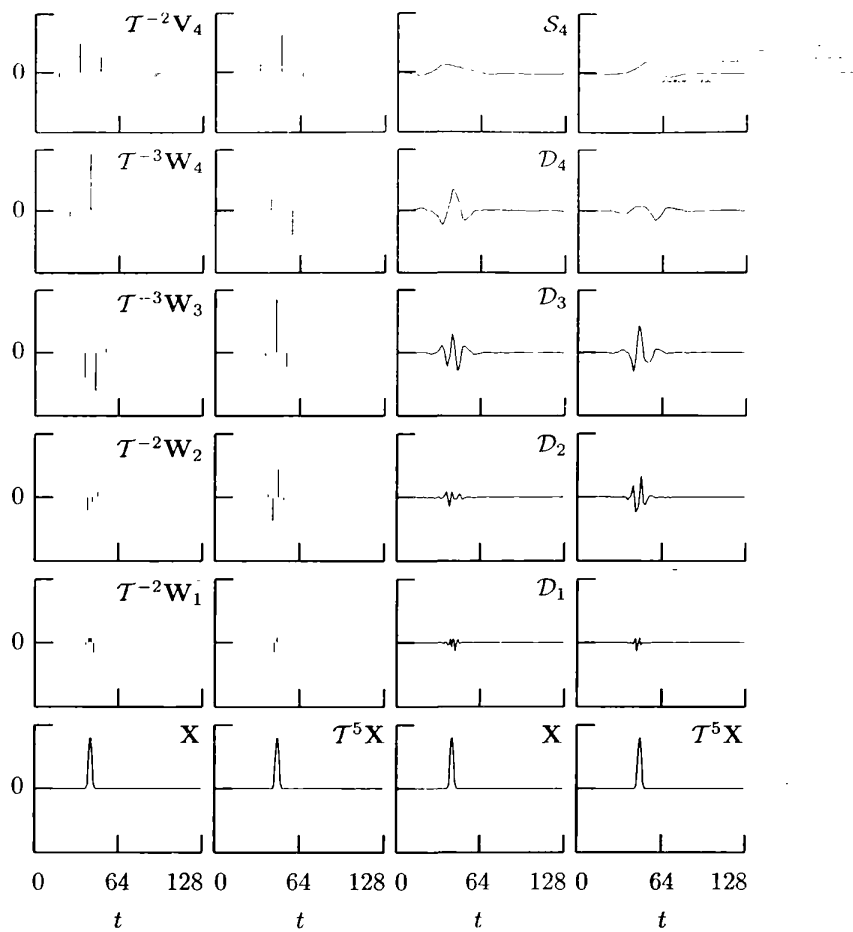


Figure 161. LA(8) DWTs of level $J_0 = 4$ (top five rows, first two columns) and corresponding multiresolution analyses (last two columns) for a time series $\mathbf{X}$ and its circularly shift $T^5\mathbf{X}$ (bottom row of plots). A comparison of the first and second columns shows that circularly shifting a time series can yield substantial changes in its DWT; likewise, the third and fourth columns indicate the same is true for the corresponding MRAs. This figure should be compared with Figure 181, which uses the MODWT in place of the DWT.

plotted as deviations from the zero line, any coefficients that are identically zero show up as blank space (there are quite a few such coefficients due to the localized natures of the bump and the DWT). Although the sum of squares of all the DWT coefficients for either $\mathbf{X}$ and $T^5\mathbf{X}$ is equal to $\|\mathbf{X}\|^2$, a comparison of the first two columns shows that shifting the time origin of the bump has altered the DWT considerably. An examination of the last two columns shows that the same is true for the MRAs. Although in each case the summation of the details and smooth yields the time series in the bottom row, corresponding details for $\mathbf{X}$ and $T^5\mathbf{X}$ can be quite different.

To understand why shifting the time origin yields such different DWTs, let us concentrate on elements $W_{4,j}$, $j = 4, \dots, 7$, of the subvectors $\mathbf{W}_4$ for $\mathbf{X}$ and $T^5\mathbf{X}$

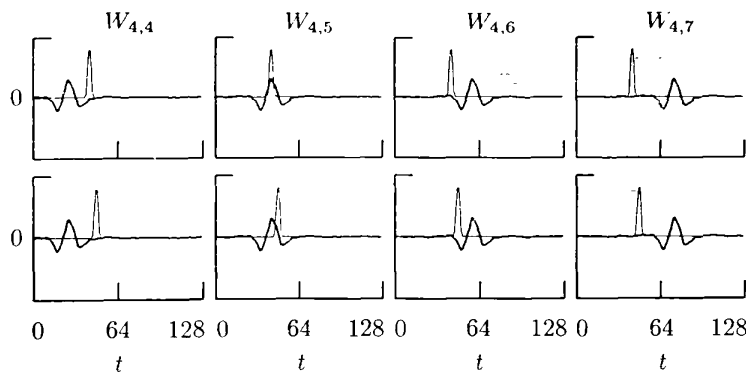


Figure 162. Level $J_0 = 4$ basis vectors used in the LA(8) DWT to produce the wavelet coefficients $W_{4,j}$, $j = 4, \dots, 7$ (thick curves, respectively, from left to right in both rows of plots), along with a ‘bump’ time series $\mathbf{X}$ (thin curves in top row of plots) and $\mathcal{T}^5 \mathbf{X}$ (thin curves, bottom row). The inner product between a plotted basis vector and a time series yields the wavelet coefficient labeled at the top of the figure.

(these four coefficients are shown in the first two plots in the second row of Figure 161 just to the right of the tick mark for the zero point on the vertical axis). These coefficients are formed by taking the inner product of the time series with a vector whose elements are in a row of the DWT matrix $\mathcal{W}$. The four row vectors are plotted as thick curves in the top and bottom rows of Figure 162, along with (as thin curves) $\mathbf{X}$ in the top row and $\mathcal{T}^5 \mathbf{X}$ in the bottom row. If we take any plot in the top row of this figure and compute the inner product of the two plotted vectors, we obtain a $W_{4,j}$ coefficient for $\mathbf{X}$; in a similar manner, the bottom row yields the corresponding coefficients for $\mathcal{T}^5 \mathbf{X}$. For either $\mathbf{X}$ or $\mathcal{T}^5 \mathbf{X}$, each level $j = 4$ LA(8) basis vector is extracting a difference between two weighted averages. The first average spans a scale of $\tau_4 = 2^3 = 8$ and is determined by the position of the positive central lobe of the basis vector. The second average involves points before and after the first average and is determined by the positions of the two negative sidelobes. If we consider the first row of Figure 162, we see that $W_{4,5}$ is large because the bump in $\mathbf{X}$ is contained wholly within the central lobe of the LA(8) basis vector, whereas the two negative sidelobes cover portions of $\mathbf{X}$ that are zero; on the other hand, $W_{4,4}$, $W_{4,6}$ and $W_{4,7}$ are all close to zero because the bump is not well aligned with either the central lobe or the sidelobes. The second row of Figure 162 shows that shifting the bump to obtain $\mathcal{T}^5 \mathbf{X}$ changes the nature of $W_{4,5}$ and $W_{4,6}$. For $W_{4,5}$, the shifted bump covers part of the central lobe and part of one sidelobe, so the resulting coefficient is much smaller than in the case of $\mathbf{X}$; on the other hand, for $W_{4,6}$, the shifted bump now matches up with one of the sidelobes, resulting in a negative coefficient.

In summary, while the wavelet coefficients for the DWT can be interpreted as a difference between two weighted averages, the intervals over which these averages are made are rigidly fixed *a priori* and hence might not line up well with interesting features in a time series. A change in the starting point for a time series can thus yield quite different results due to the juxtaposition of the time series with the averaging intervals predefined by the DWT. As we shall see, the MODWT is an attempt to get away from effects attributable to the choice of a starting time by essentially including

all possible placements of averaging intervals.

5.2 MODWT Wavelet and Scaling Filters

In order to easily make connections between the DWT and the MODWT when we formulate the latter in the next two sections, it is convenient to define a MODWT wavelet filter $\{\tilde{h}_l\}$ via $\tilde{h}_l \equiv h_l/\sqrt{2}$ and a MODWT scaling filter $\{\tilde{g}_l\}$ via $\tilde{g}_l \equiv g_l/\sqrt{2}$. As $\{\tilde{h}_l\}$ is simply a rescaled version of the wavelet filter, it readily follows from Equations (69a), (69b) and (69c) that

$$\sum_{l=0}^{L-1} \tilde{h}_l = 0, \quad \sum_{l=0}^{L-1} \tilde{h}_l^2 = \frac{1}{2} \quad \text{and} \quad \sum_{l=-\infty}^{\infty} \tilde{h}_l \tilde{h}_{l+2n} = 0 \quad (163a)$$

for all nonzero integers n . If we let $\tilde{H}(\cdot)$ and $\tilde{\mathcal{H}}(\cdot)$ denote, respectively, the transfer and squared gain functions for $\{\tilde{h}_l\}$, we have $\tilde{H}(f) = H(f)/\sqrt{2}$ and $\tilde{\mathcal{H}}(f) = \mathcal{H}(f)/2$, so Equation (69d) leads to the relationship

$$\tilde{\mathcal{H}}(f) + \tilde{\mathcal{H}}(f + \frac{1}{2}) = 1 \quad \text{for all } f. \quad (163b)$$

Similarly, for the MODWT scaling filter, it follows from Exercise [76b] that

$$\sum_{l=0}^{L-1} \tilde{g}_l = 1, \quad \sum_{l=0}^{L-1} \tilde{g}_l^2 = \frac{1}{2} \quad \text{and} \quad \sum_{l=-\infty}^{\infty} \tilde{g}_l \tilde{g}_{l+2n} = 0$$

for all nonzero integers n , while it follows from Exercise [77] that

$$\sum_{l=-\infty}^{\infty} \tilde{g}_l \tilde{h}_{l+2n} = 0$$

for all integers n . If we let $\tilde{G}(\cdot)$ and $\tilde{\mathcal{G}}(\cdot)$ denote, respectively, the transfer and squared gain functions for $\{\tilde{g}_l\}$, we have $\tilde{G}(f) = G(f)/\sqrt{2}$ and $\tilde{\mathcal{G}}(f) = \mathcal{G}(f)/2$, so Equation (76b) tells us that

$$\tilde{\mathcal{G}}(f) + \tilde{\mathcal{G}}(f + \frac{1}{2}) = 1 \quad \text{and} \quad \tilde{\mathcal{G}}(f) + \tilde{\mathcal{H}}(f) = 1 \quad \text{for all } f. \quad (163c)$$

Note also that the quadrature mirror relationships of Equations (75a) and (75b) also hold for the MODWT filters:

$$\tilde{g}_l = (-1)^{l+1} \tilde{h}_{L-1-l} \quad \text{and} \quad \tilde{h}_l = (-1)^l \tilde{g}_{L-1-l}. \quad (163d)$$

Using the above, we can re-express Equations (70b) and (77a) as

$$\tilde{W}_{1,t} \equiv \sum_{l=0}^{L-1} \tilde{h}_l X_{t-l \bmod N} \quad \text{and} \quad \tilde{V}_{1,t} \equiv \sum_{l=0}^{L-1} \tilde{g}_l X_{t-l \bmod N}, \quad (163e)$$

$t = 0, \dots, N-1$. Thus the sequences $\{\tilde{W}_{1,t}\}$ and $\{\tilde{V}_{1,t}\}$ are obtained by circularly filtering $\{X_t\}$ with, respectively, the MODWT wavelet and scaling filters. (As we shall see in the following section, these two sequences in fact constitute the MODWT of level $J_0 = 1$.)

5.3 Basic Concepts for MODWT

The motivation for formulating the MODWT is essentially to define a transform that acts as much as possible like the DWT, but does not suffer from the DWT's sensitivity to the choice of a starting point for a time series. This sensitivity is entirely due to downsampling (subsampling) the outputs from the wavelet and scaling filters at each stage of the pyramid algorithm. To obtain a degree of insensitivity to the starting point, we need somehow to eliminate this downsampling while preserving the ability to carry out an ANOVA and an MRA.

As a first step in this direction, note that the filter outputs that are usually discarded at the first stage of the DWT pyramid algorithm can be obtained by applying the DWT pyramid algorithm to the circularly shifted vector $\mathcal{T}\mathbf{X}$ rather than $\mathbf{X}$. This suggests the following procedure for defining the first stage of the pyramid algorithm for the MODWT when N is an even sample size (an assumption that we will in fact drop later on). The idea is to apply the usual DWT pyramid algorithm twice, once to $\mathbf{X}$ and once to $\mathcal{T}\mathbf{X}$, after which we merge the two sets of DWT coefficients together. The first application yields the usual DWT, namely,

$$\begin{bmatrix} \mathbf{W}_1 \\ \mathbf{V}_1 \end{bmatrix} = \mathcal{P}_1 \mathbf{X} = \begin{bmatrix} \mathcal{B}_1 \\ \mathcal{A}_1 \end{bmatrix} \mathbf{X},$$

where (appealing to Equation (81b) since $\mathcal{W}_1 = \mathcal{B}_1$)

$$\mathcal{B}_1 = \begin{bmatrix} h_1 & h_0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & h_3 & h_2 \\ h_3 & h_2 & h_1 & h_0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & h_3 & h_2 & h_1 & h_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & h_3 & h_2 & h_1 & h_0 \end{bmatrix},$$

and $\mathcal{A}_1$ has the same structure as the above with each h_l replaced by g_l (in displaying the elements of various matrices in this section, we specialize to the case $L = 4$ and $N > L$ for clarity, but our mathematical treatment holds in general). In view of Equations (70c) and (77b), we can denote the elements of $\mathbf{W}_1$ and $\mathbf{V}_1$ by

$$\mathbf{W}_1 = [2^{1/2}\widetilde{W}_{1,1}, 2^{1/2}\widetilde{W}_{1,3}, \dots, 2^{1/2}\widetilde{W}_{1,N-1}]^T$$

and

$$\mathbf{V}_1 = [2^{1/2}\widetilde{V}_{1,1}, 2^{1/2}\widetilde{V}_{1,3}, \dots, 2^{1/2}\widetilde{V}_{1,N-1}]^T;$$

i.e., $\mathbf{W}_1$ and $\mathbf{V}_1$ contain all the odd indexed elements of the length N sequences $\{2^{1/2}\widetilde{W}_{1,t}\}$ and $\{2^{1/2}\widetilde{V}_{1,t}\}$, which are formed by circularly convolving the time series $\mathbf{X}$ with, respectively, the wavelet filter $\{h_l\}$ and the scaling filter $\{g_l\}$. The second application consists of substituting the circularly shifted vector $\mathcal{T}\mathbf{X}$ for $\mathbf{X}$ to obtain

$$\begin{bmatrix} \mathbf{W}_{\mathcal{T},1} \\ \mathbf{V}_{\mathcal{T},1} \end{bmatrix} \equiv \mathcal{P}_1 \mathcal{T}\mathbf{X}.$$

Equivalently, if we define

$$\mathcal{P}_{\mathcal{T},1} \equiv \mathcal{P}_1 \mathcal{T} = \begin{bmatrix} \mathcal{B}_1 \\ \mathcal{A}_1 \end{bmatrix} \mathcal{T} = \begin{bmatrix} \mathcal{B}_1 \mathcal{T} \\ \mathcal{A}_1 \mathcal{T} \end{bmatrix} \equiv \begin{bmatrix} \mathcal{B}_{\mathcal{T},1} \\ \mathcal{A}_{\mathcal{T},1} \end{bmatrix},$$

we can write

$$\begin{bmatrix} \mathbf{W}_{T,1} \\ \mathbf{V}_{T,1} \end{bmatrix} = \mathcal{P}_{T,1} \mathbf{X},$$

where

$$\mathcal{B}_{T,1} = \begin{bmatrix} h_3 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & h_3 & h_2 & h_1 \\ h_2 & h_1 & h_0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 & 0 & h_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & h_3 & h_2 & h_1 & h_0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & h_3 & h_2 & h_1 & h_0 & 0 & 0 \end{bmatrix},$$

and $\mathcal{A}_{T,1}$ has a similar structure with each h_i being replaced by g_i . A comparison of the contents of $\mathcal{B}_1$ and $\mathcal{B}_{T,1}$ says that, because $\mathcal{B}_1 \mathbf{X}$ forms the odd indexed values of the filter output $\{2^{1/2} \widetilde{W}_{1,t}\}$, then $\mathcal{B}_{T,1} \mathbf{X}$ must form the even indexed values. Hence the elements of $\mathbf{W}_{T,1}$ are given by

$$\mathbf{W}_{T,1} = [2^{1/2} \widetilde{W}_{1,0}, 2^{1/2} \widetilde{W}_{1,2}, \dots, 2^{1/2} \widetilde{W}_{1,N-2}]^T,$$

and by a similar argument those of $\mathbf{V}_{T,1}$ are given by

$$\mathbf{V}_{T,1} = [2^{1/2} \widetilde{V}_{1,0}, 2^{1/2} \widetilde{V}_{1,2}, \dots, 2^{1/2} \widetilde{V}_{1,N-2}]^T$$

Next we define

$$\widetilde{\mathbf{W}}_1 \equiv [\widetilde{W}_{1,0}, \widetilde{W}_{1,1}, \widetilde{W}_{1,2}, \dots, \widetilde{W}_{1,N-1}]^T$$

and

$$\widetilde{\mathbf{V}}_1 \equiv [\widetilde{V}_{1,0}, \widetilde{V}_{1,1}, \widetilde{V}_{1,2}, \dots, \widetilde{V}_{1,N-1}]^T;$$

i.e., $\widetilde{\mathbf{W}}_1$ is formed by rescaling the interleaved elements of $\mathbf{W}_1$ and $\mathbf{W}_{T,1}$, and $\widetilde{\mathbf{V}}_1$ is constructed in a similar fashion from $\mathbf{V}_1$ and $\mathbf{V}_{T,1}$. Note that the elements of $\widetilde{\mathbf{W}}_1$ and $\widetilde{\mathbf{V}}_1$ are exactly the filter outputs $\{\widetilde{W}_{1,t}\}$ and $\{\widetilde{V}_{1,t}\}$ obtained by using the MODWT filters $\{\tilde{h}_l\}$ and $\{\tilde{g}_l\}$ – see Equation (163e). Let us define $\tilde{\mathcal{B}}_1$ as the $N \times N$ matrix formed by interleaving the rows of $\mathcal{B}_{T,1}$ and $\mathcal{B}_1$ and replacing each h_l by $\tilde{h}_l$; i.e.,

$$\tilde{\mathcal{B}}_1 \equiv \begin{bmatrix} \tilde{h}_0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \tilde{h}_3 & \tilde{h}_2 & \tilde{h}_1 \\ \tilde{h}_1 & \tilde{h}_0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & \tilde{h}_3 & \tilde{h}_2 \\ \tilde{h}_2 & \tilde{h}_1 & \tilde{h}_0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 & \tilde{h}_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & \tilde{h}_3 & \tilde{h}_2 & \tilde{h}_1 & \tilde{h}_0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & \tilde{h}_3 & \tilde{h}_2 & \tilde{h}_1 & \tilde{h}_0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \tilde{h}_3 & \tilde{h}_2 & \tilde{h}_1 & \tilde{h}_0 \end{bmatrix}. \quad (165a)$$

We then have $\widetilde{\mathbf{W}}_1 = \tilde{\mathcal{B}}_1 \mathbf{X}$. With an analogous definition for $\tilde{\mathcal{A}}_1$, we have $\widetilde{\mathbf{V}}_1 = \tilde{\mathcal{A}}_1 \mathbf{X}$. Finally, we can represent the first stage of the MODWT pyramid algorithm as

$$\begin{bmatrix} \widetilde{\mathbf{W}}_1 \\ \widetilde{\mathbf{V}}_1 \end{bmatrix} = \begin{bmatrix} \tilde{\mathcal{B}}_1 \\ \tilde{\mathcal{A}}_1 \end{bmatrix} \mathbf{X} = \tilde{\mathcal{P}}_1 \mathbf{X} \quad \text{with} \quad \tilde{\mathcal{P}}_1 \equiv \begin{bmatrix} \tilde{\mathcal{B}}_1 \\ \tilde{\mathcal{A}}_1 \end{bmatrix}. \quad (165b)$$

Because $\mathcal{P}_1^T \mathcal{P}_1 = I_N$ and $\mathcal{T}^T \mathcal{T} = I_N$, it follows that

$$\mathcal{P}_{\mathcal{T},1}^T \mathcal{P}_{\mathcal{T},1} = \mathcal{T}^T \mathcal{P}_1^T \mathcal{P}_1 \mathcal{T} = I_N$$

and hence that $\mathcal{P}_{\mathcal{T},1}$ is an orthonormal matrix. Hence we have both

$$\|\mathbf{X}\|^2 = \|\mathbf{W}_1\|^2 + \|\mathbf{V}_1\|^2 \text{ and } \|\mathbf{X}\|^2 = \|\mathbf{W}_{\mathcal{T},1}\|^2 + \|\mathbf{V}_{\mathcal{T},1}\|^2.$$

Since

$$\|\mathbf{W}_1\|^2 + \|\mathbf{W}_{\mathcal{T},1}\|^2 = 2\|\widetilde{\mathbf{W}}_1\|^2 \text{ and } \|\mathbf{V}_1\|^2 + \|\mathbf{V}_{\mathcal{T},1}\|^2 = 2\|\widetilde{\mathbf{V}}_1\|^2,$$

it follows that

$$\|\mathbf{X}\|^2 = \|\widetilde{\mathbf{W}}_1\|^2 + \|\widetilde{\mathbf{V}}_1\|^2,$$

so that $\|\widetilde{\mathbf{W}}_1\|^2$ and $\|\widetilde{\mathbf{V}}_1\|^2$ decompose $\|\mathbf{X}\|^2$, which is the basic result needed to define an ANOVA using the MODWT.

Since we have both

$$\mathbf{X} = [\mathcal{B}_1^T, \mathcal{A}_1^T] \begin{bmatrix} \mathbf{W}_1 \\ \mathbf{V}_1 \end{bmatrix} \text{ and } \mathbf{X} = [\mathcal{B}_{\mathcal{T},1}^T, \mathcal{A}_{\mathcal{T},1}^T] \begin{bmatrix} \mathbf{W}_{\mathcal{T},1} \\ \mathbf{V}_{\mathcal{T},1} \end{bmatrix},$$

we can also recover $\mathbf{X}$ by averaging the two right-hand sides:

$$\begin{aligned} \mathbf{X} &= \frac{1}{2} [\mathcal{B}_1^T, \mathcal{A}_1^T] \begin{bmatrix} \mathbf{W}_1 \\ \mathbf{V}_1 \end{bmatrix} + \frac{1}{2} [\mathcal{B}_{\mathcal{T},1}^T, \mathcal{A}_{\mathcal{T},1}^T] \begin{bmatrix} \mathbf{W}_{\mathcal{T},1} \\ \mathbf{V}_{\mathcal{T},1} \end{bmatrix} \\ &= \frac{1}{2} (\mathcal{B}_1^T \mathbf{W}_1 + \mathcal{A}_1^T \mathbf{V}_1 + \mathcal{B}_{\mathcal{T},1}^T \mathbf{W}_{\mathcal{T},1} + \mathcal{A}_{\mathcal{T},1}^T \mathbf{V}_{\mathcal{T},1}). \end{aligned}$$

▷ **Exercise [166]:** Argue that

$$\frac{1}{2} (\mathcal{B}_1^T \mathbf{W}_1 + \mathcal{B}_{\mathcal{T},1}^T \mathbf{W}_{\mathcal{T},1}) = \widetilde{\mathcal{B}}_1^T \widetilde{\mathbf{W}}_1$$

and

$$\frac{1}{2} (\mathcal{A}_1^T \mathbf{V}_1 + \mathcal{A}_{\mathcal{T},1}^T \mathbf{V}_{\mathcal{T},1}) = \widetilde{\mathcal{A}}_1^T \widetilde{\mathbf{V}}_1.$$

With the above, we can write

$$\mathbf{X} = \widetilde{\mathcal{B}}_1^T \widetilde{\mathbf{W}}_1 + \widetilde{\mathcal{A}}_1^T \widetilde{\mathbf{V}}_1 \equiv \widetilde{\mathcal{D}}_1 + \widetilde{\mathcal{S}}_1,$$

where $\widetilde{\mathcal{D}}_1 \equiv \widetilde{\mathcal{B}}_1^T \widetilde{\mathbf{W}}_1$ is the first level maximal overlap detail, while $\widetilde{\mathcal{S}}_1 \equiv \widetilde{\mathcal{A}}_1^T \widetilde{\mathbf{V}}_1$ is the corresponding smooth. The above equation is the basic result needed to define an MRA of level $J_0 = 1$ using the MODWT. (Exercise [5.1] explores an interesting interpretation of $\widetilde{\mathcal{D}}_1$ and $\widetilde{\mathcal{S}}_1$. We can in fact obtain $\widetilde{\mathcal{D}}_1$ by averaging $\mathcal{D}_1$ and $\mathcal{T}\mathcal{D}_{\mathcal{T},1}$, where $\mathcal{D}_1$ and $\mathcal{D}_{\mathcal{T},1}$ are, respectively, first level DWT details for $\mathbf{X}$ and $\mathcal{T}\mathbf{X}$. An analogous statement holds for $\widetilde{\mathcal{S}}_1$.)

It is important to note that, whereas we have the relationship $\|\mathbf{W}_1\|^2 = \|\mathcal{D}_1\|^2$ for the usual DWT, we do *not* in general have equality between $\|\widetilde{\mathbf{W}}_1\|^2$ and $\|\widetilde{\mathcal{D}}_1\|^2$, as the following exercise indicates.

▷ **Exercise [167a]:** Show that

$$\|\tilde{\mathcal{D}}_1\|^2 = \frac{1}{2} \left(\|\tilde{\mathbf{W}}_1\|^2 + \mathbf{W}_1^T \mathcal{B}_1 \mathcal{B}_1^T \mathbf{W}_{T,1} \right).$$

Thus, whereas we can define the discrete wavelet empirical power spectrum at scale τ_1 using the usual DWT as either $\|\mathbf{W}_1\|^2/N$ or $\|\mathcal{D}_1\|^2/N$, we can only use $\|\tilde{\mathbf{W}}_1\|^2/N$ when dealing with the MODWT. (Exercise [5.7] indicates that in fact $\|\tilde{\mathcal{D}}_1\|^2 \leq \|\tilde{\mathbf{W}}_1\|^2$, but this exercise is best tackled after studying Section 5.4.)

The above derivation of the MODWT pyramid algorithm assumes that N is even so that we can apply the usual DWT pyramid algorithm to $\mathbf{X}$ and to $\mathcal{T}\mathbf{X}$. The following exercise says that the first stage of this algorithm in fact works for all N .

▷ **Exercise [167b]:** For any sample size $N \geq L$, let $\tilde{\mathcal{B}}_1$ be defined as in Equation (165a), and define $\tilde{\mathcal{A}}_1$ analogously by substituting $\tilde{g}_l$'s for $\tilde{h}_l$'s. Suppose that $\tilde{\mathbf{W}}_1$ and $\tilde{\mathbf{V}}_1$ are defined by Equation (165b). Show that

$$\mathbf{X} = \tilde{\mathcal{B}}_1^T \tilde{\mathbf{W}}_1 + \tilde{\mathcal{A}}_1^T \tilde{\mathbf{V}}_1 \quad \text{and} \quad \|\mathbf{X}\|^2 = \|\tilde{\mathbf{W}}_1\|^2 + \|\tilde{\mathbf{V}}_1\|^2.$$

Hint: show that $\tilde{\mathcal{B}}_1^T \tilde{\mathcal{B}}_1 + \tilde{\mathcal{A}}_1^T \tilde{\mathcal{A}}_1 = I_N$.

In fact, as indicated by the discussion in Section 5.4, the restriction $N \geq L$ can be relaxed.

With $\tilde{\mathcal{P}}_1$ defined as in Equation (165b), we can re-express $\mathbf{X} = \tilde{\mathcal{B}}_1^T \tilde{\mathbf{W}}_1 + \tilde{\mathcal{A}}_1^T \tilde{\mathbf{V}}_1$ as

$$\mathbf{X} = \begin{bmatrix} \tilde{\mathcal{B}}_1 \\ \tilde{\mathcal{A}}_1 \end{bmatrix}^T \begin{bmatrix} \tilde{\mathbf{W}}_1 \\ \tilde{\mathbf{V}}_1 \end{bmatrix} = \tilde{\mathcal{P}}_1^T \begin{bmatrix} \tilde{\mathbf{W}}_1 \\ \tilde{\mathbf{V}}_1 \end{bmatrix}; \quad \text{furthermore, since} \quad \begin{bmatrix} \tilde{\mathbf{W}}_1 \\ \tilde{\mathbf{V}}_1 \end{bmatrix} = \tilde{\mathcal{P}}_1 \mathbf{X},$$

it is evident that $\mathbf{X} = \tilde{\mathcal{P}}_1^T \tilde{\mathcal{P}}_1 \mathbf{X}$ for any time series and hence that $\tilde{\mathcal{P}}_1^T$ can be considered as an inverse to the MODWT transform $\tilde{\mathcal{P}}_1$. Because $\tilde{\mathcal{P}}_1$ is a $2N \times N$ matrix, there are in fact an infinite number of ways of recovering $\mathbf{X}$ from its MODWT coefficients $\tilde{\mathbf{W}}_1$ and $\tilde{\mathbf{V}}_1$, so $\tilde{\mathcal{P}}_1^T$ is by no means a unique inverse for $\tilde{\mathcal{P}}_1$; however, Exercise [5.2] is to show that $\tilde{\mathcal{P}}_1^T$ is in fact the (unique) Moore–Penrose generalized inverse for $\tilde{\mathcal{P}}_1$.

For the DWT, we argued in Section 4.4 that the detail $\mathcal{D}_1$ can be constructed by cross-correlating the periodized filter $\{h_l^\circ\}$ with an upsampled version of $\mathbf{W}_1$. Here we show that the MODWT detail $\tilde{\mathcal{D}}_1$ can be interpreted in terms of a filtering operation involving $\mathbf{X}$ and a zero phase filter. Let $\tilde{\mathcal{D}}_{1,t}$ be the t th element of $\tilde{\mathcal{D}}_1$. Expanding out $\tilde{\mathcal{D}}_1 = \tilde{\mathcal{B}}_1^T \tilde{\mathbf{W}}_1$ yields

$$\tilde{\mathcal{D}}_1 = \begin{bmatrix} \tilde{h}_0 & \tilde{h}_1 & \tilde{h}_2 & \tilde{h}_3 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & \tilde{h}_0 & \tilde{h}_1 & \tilde{h}_2 & \tilde{h}_3 & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & \tilde{h}_0 & \tilde{h}_1 & \tilde{h}_2 & \tilde{h}_3 \\ \tilde{h}_3 & 0 & 0 & 0 & 0 & \cdots & 0 & \tilde{h}_0 & \tilde{h}_1 & \tilde{h}_2 \\ \tilde{h}_2 & \tilde{h}_3 & 0 & 0 & 0 & \cdots & 0 & 0 & \tilde{h}_0 & \tilde{h}_1 \\ \tilde{h}_1 & \tilde{h}_2 & \tilde{h}_3 & 0 & 0 & \cdots & 0 & 0 & 0 & \tilde{h}_0 \end{bmatrix} \begin{bmatrix} \tilde{W}_{1,0} \\ \tilde{W}_{1,1} \\ \tilde{W}_{1,2} \\ \vdots \\ \tilde{W}_{1,N-2} \\ \tilde{W}_{1,N-1} \end{bmatrix},$$

from which we have

$$\tilde{\mathcal{D}}_{1,t} = \sum_{l=0}^{L-1} \tilde{h}_l \tilde{W}_{1,t+l \bmod N} = \sum_{l=0}^{N-1} \tilde{h}_l^\circ \tilde{W}_{1,t+l \bmod N}, \quad t = 0, 1, \dots, N-1.$$

where $\{\tilde{h}_l^\circ : l = 0, \dots, N-1\}$ is $\{\tilde{h}_l : l = 0, \dots, L-1\}$ periodized to length N . Thus the detail series can be obtained by circularly cross-correlating $\{\tilde{W}_{1,t}\}$ with $\{\tilde{h}_l^\circ\}$ (cf. Equation (37c)). Since $\tilde{H}(\cdot)$ is the transfer function for $\{\tilde{h}_l\}$, an equivalent statement is that the detail series can be obtained by circularly filtering $\{\tilde{W}_{1,t}\}$ with a filter whose DFT is $\{\tilde{H}^*(\frac{k}{N}) : k = 0, \dots, N-1\}$. Since $\{\tilde{W}_{1,t}\}$ was formed by filtering $\{X_t\}$ using $\{\tilde{h}_l^\circ\}$, the detail series can equivalently be formed by filtering $\{X_t\}$ with a composite filter obtained by convolving $\{\tilde{h}_l^\circ\}$ with the filter whose DFT is $\{\tilde{H}^*(\frac{k}{N})\}$. Now the DFT for the composite circular filter is given by the product of the DFTs for the individual filters, which yields

$$\tilde{H}^*(\frac{k}{N})\tilde{H}(\frac{k}{N}) = |\tilde{H}(\frac{k}{N})|^2 = \tilde{\mathcal{H}}(\frac{k}{N}).$$

Because $\tilde{\mathcal{H}}(\cdot)$ is real-valued and nonnegative, its phase function is zero for all frequencies (if we define it as such for $f = 0$, where $\tilde{\mathcal{H}}(0) = 0$), which establishes the important result that the composite filter associated with $\tilde{\mathcal{D}}_1$ has zero phase. Using a similar argument, it follows that $\tilde{\mathcal{S}}_1$ can be expressed as filtering $\{\tilde{V}_{1,t}\}$ with a filter whose DFT is $\{\tilde{G}^*(\frac{k}{N})\}$, where $\tilde{G}(\cdot)$ is the transfer function for $\{\tilde{g}_l\}$. The composite filter we would apply to $\mathbf{X}$ to obtain $\tilde{\mathcal{S}}_1$ has a DFT composed of samples from a zero phase transfer function defined by $\tilde{\mathcal{G}}(f) \equiv |\tilde{G}(f)|^2$.

Key Facts and Definitions in Section 5.3

To summarize, the N dimensional vector $\tilde{\mathbf{W}}_1$ contains the MODWT wavelet coefficients of level $j = 1$. The t th element of $\tilde{\mathbf{W}}_1$ is $\tilde{W}_{1,t}$, $t = 0, 1, \dots, N-1$, and is formed by circularly filtering the time series $\{X_t\}$ with the MODWT wavelet filter $\{\tilde{h}_l \equiv h_l/\sqrt{2}\}$ (see Equation (163e)); equivalently, we can regard $\{\tilde{W}_{1,t}\}$ as the result of circularly filtering $\{X_t\}$ with the filter $\{\tilde{h}_l^\circ\}$, which is $\{\tilde{h}_l\}$ periodized to length N . Likewise, the MODWT scaling coefficients $\tilde{\mathbf{V}}_1$ are obtained by filtering $\{X_t\}$ with the MODWT scaling filter $\{\tilde{g}_l \equiv g_l/\sqrt{2}\}$; equivalently, we can consider $\{\tilde{V}_{1,t}\}$ to be the result of filtering $\{X_t\}$ with $\{\tilde{g}_l^\circ\}$, which is the periodized version of $\{\tilde{g}_l\}$. Each row of the $N \times N$ matrix $\tilde{\mathcal{B}}_1$ contains the periodized filter $\{\tilde{h}_l^\circ\}$, with the difference between rows given by circular shifts; likewise, $\tilde{\mathcal{A}}_1$ has rows whose elements are the periodized filter $\{\tilde{g}_l^\circ\}$. With these definitions, we have

$$\|\mathbf{X}\|^2 = \|\tilde{\mathbf{W}}_1\|^2 + \|\tilde{\mathbf{V}}_1\|^2 \text{ and } \mathbf{X} = \tilde{\mathcal{B}}_1^T \tilde{\mathbf{W}}_1 + \tilde{\mathcal{A}}_1^T \tilde{\mathbf{V}}_1 \equiv \tilde{\mathcal{D}}_1 + \tilde{\mathcal{S}}_1,$$

which yield, respectively, an energy decomposition and an additive decomposition in terms of the first level MODWT detail $\tilde{\mathcal{D}}_1$ and the first level MODWT smooth $\tilde{\mathcal{S}}_1$. The elements of $\tilde{\mathcal{D}}_1$ can be obtained by filtering $\tilde{\mathbf{W}}_1$ with a filter whose transfer function is $\tilde{H}^*(\cdot)$, i.e., the complex conjugate of the transfer function $\tilde{H}(\cdot)$ of $\{\tilde{h}_l\}$; likewise, the elements of $\tilde{\mathcal{S}}_1$ are the result of filtering $\tilde{\mathbf{V}}_1$ with a filter whose transfer function is $\tilde{G}^*(\cdot)$, which is the complex conjugate of the transfer function for $\{\tilde{g}_l\}$. A flow diagram depicting the analysis and synthesis of $\mathbf{X}$ in terms of MODWT coefficients is shown in Figure 169.

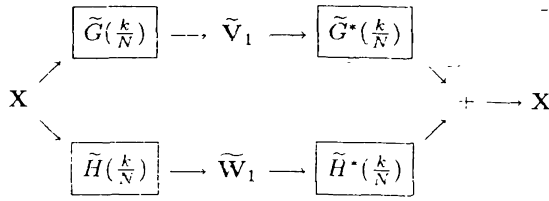


Figure 169. Flow diagram illustrating analysis of $\mathbf{X}$ into the MODWT wavelet and scaling coefficients $\tilde{\mathbf{W}}_1$ and $\tilde{\mathbf{V}}_1$ of first level, followed by the synthesis of $\mathbf{X}$ from $\tilde{\mathbf{W}}_1$ and $\tilde{\mathbf{V}}_1$.

5.4 Definition of j th Level MODWT Coefficients

For arbitrary sample size N , we now *define* the j th-level MODWT wavelet and scaling coefficients to be the N dimensional vectors $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ whose elements are, respectively,

$$\tilde{W}_{j,t} \equiv \sum_{l=0}^{L_j-1} \tilde{h}_{j,l} X_{t-l \bmod N} \quad \text{and} \quad \tilde{V}_{j,t} \equiv \sum_{l=0}^{L_j-1} \tilde{g}_{j,l} X_{t-l \bmod N}, \quad (169a)$$

$t = 0, \dots, N-1$, where $\tilde{h}_{j,l} \equiv h_{j,l}/2^{j/2}$ and $\tilde{g}_{j,l} \equiv g_{j,l}/2^{j/2}$. The filters $\{\tilde{h}_{j,l}\}$ and $\{\tilde{g}_{j,l}\}$ are called the j th level MODWT wavelet and scaling filters. Note that, because $\{h_{j,l}\}$ and $\{g_{j,l}\}$ each have width

$$L_j \equiv (2^j - 1)(L - 1) + 1$$

(cf. Exercise [96]), so do $\{\tilde{h}_{j,l}\}$ and $\{\tilde{g}_{j,l}\}$. Because the transfer functions for $\{\tilde{h}_l\}$ and $\{\tilde{g}_l\}$ are given by $\tilde{H}(f) \equiv H(f)/\sqrt{2}$ and $\tilde{G}(f) \equiv G(f)/\sqrt{2}$, it is evident from Equations (96b) and (97) that the transfer functions for $\{\tilde{h}_{j,l}\}$ and $\{\tilde{g}_{j,l}\}$ are given, respectively, by

$$\tilde{H}_j(f) \equiv \tilde{H}(2^{j-1}f) \prod_{l=0}^{j-2} \tilde{G}(2^l f) \quad \text{and} \quad \tilde{G}_j(f) \equiv \prod_{l=0}^{j-1} \tilde{G}(2^l f). \quad (169b)$$

For $j = 1$, the above definitions yield $L_1 \equiv L$ and $\tilde{G}_1(f) \equiv \tilde{G}(f)$; we also take $\tilde{H}_1(f) \equiv \tilde{H}(f)$, $\tilde{h}_{1,l} \equiv \tilde{h}_l$ and $\tilde{g}_{1,l} \equiv \tilde{g}_l$.

We now have two tasks in front of us. First, given any integer $J_0 \geq 1$ and any sample size N , we need to prove that our definition for the MODWT yields an energy decomposition

$$\|\mathbf{X}\|^2 = \sum_{j=1}^{J_0} \|\tilde{\mathbf{W}}_j\|^2 + \|\tilde{\mathbf{V}}_{J_0}\|^2 \quad (169c)$$

and an additive decomposition

$$\mathbf{X} = \sum_{j=1}^{J_0} \tilde{\mathcal{D}}_j + \tilde{\mathcal{S}}_{J_0}, \quad (169d)$$

where $\tilde{\mathcal{D}}_j$ and $\tilde{\mathcal{S}}_{J_0}$ depend on $\mathbf{X}$ only through, respectively, $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_{J_0}$ in a yet to be defined way. Note that the above results are similar to the ones derived in Section 4.7

for the DWT (see Equations (104b) and (104a)). Second, we need to complete our description of the pyramid algorithm for the MODWT, which we described in the previous section for just the first stage. The first task is the subject of the remainder of this section, while the next section (5.5) is devoted to specification of the MODWT pyramid algorithm.

To establish Equation (169c), let $\{\tilde{h}_{j,l}^\circ\}$ and $\{\tilde{g}_{j,l}^\circ\}$ be the filters obtained by periodizing $\{\tilde{h}_{j,l}\}$ and $\{\tilde{g}_{j,l}\}$ to length N (see Section 2.5). The DFTs of $\{\tilde{h}_{j,l}^\circ\}$ and $\{\tilde{g}_{j,l}^\circ\}$ are thus given, respectively, by $\{\tilde{H}_j(\frac{k}{N})\}$ and $\{\tilde{G}_j(\frac{k}{N})\}$. Because circularly filtering $\{X_t\}$ with $\{\tilde{h}_{j,l}\}$ is equivalent to filtering it with $\{\tilde{h}_{j,l}^\circ\}$, we can re-express Equation (169a) as

$$\widetilde{W}_{j,t} = \sum_{l=0}^{N-1} \tilde{h}_{j,l}^\circ X_{t-l \bmod N} \quad \text{and} \quad \widetilde{V}_{j,t} = \sum_{l=0}^{N-1} \tilde{g}_{j,l}^\circ X_{t-l \bmod N}, \quad (170a)$$

$t = 0, \dots, N-1$. Letting $\{\mathcal{X}_k\}$ be the DFT of $\{X_t\}$, we have

$$\{\widetilde{W}_{j,t}\} \longleftrightarrow \{\tilde{H}_j(\frac{k}{N})\mathcal{X}_k\} \quad \text{and} \quad \{\widetilde{V}_{j,t}\} \longleftrightarrow \{\tilde{G}_j(\frac{k}{N})\mathcal{X}_k\} \quad (170b)$$

(see Equations (37a) and (37b)). Parseval's theorem (Equation (36h)) now tells us that

$$\|\widetilde{W}_j\|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |\tilde{H}_j(\frac{k}{N})|^2 |\mathcal{X}_k|^2 \quad \text{and} \quad \|\widetilde{V}_j\|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |\tilde{G}_j(\frac{k}{N})|^2 |\mathcal{X}_k|^2, \quad (170c)$$

which in turn yields

$$\|\widetilde{W}_j\|^2 + \|\widetilde{V}_j\|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |\mathcal{X}_k|^2 \left(|\tilde{H}_j(\frac{k}{N})|^2 + |\tilde{G}_j(\frac{k}{N})|^2 \right).$$

For any $j \geq 2$, we can use Equation (169b) to reduce the terms in the parentheses above as follows:

$$\begin{aligned} |\tilde{H}_j(\frac{k}{N})|^2 + |\tilde{G}_j(\frac{k}{N})|^2 &= |\tilde{H}(2^{j-1} \frac{k}{N})|^2 \prod_{l=0}^{j-2} |\tilde{G}(2^l \frac{k}{N})|^2 + \prod_{l=0}^{j-1} |\tilde{G}(2^l \frac{k}{N})|^2 \\ &= \left(|\tilde{H}(2^{j-1} \frac{k}{N})|^2 + |\tilde{G}(2^{j-1} \frac{k}{N})|^2 \right) \prod_{l=0}^{j-2} |\tilde{G}(2^l \frac{k}{N})|^2 \\ &= \left(\tilde{\mathcal{H}}(2^{j-1} \frac{k}{N}) + \tilde{\mathcal{G}}(2^{j-1} \frac{k}{N}) \right) |\tilde{G}_{j-1}(\frac{k}{N})|^2 \\ &= |\tilde{G}_{j-1}(\frac{k}{N})|^2, \end{aligned} \quad (170d)$$

where we have used the fact that $\tilde{\mathcal{H}}(f) + \tilde{\mathcal{G}}(f) = 1$ for all f (this is Equation (163c)). We now have

$$\|\widetilde{W}_j\|^2 + \|\widetilde{V}_j\|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |\mathcal{X}_k|^2 |\tilde{G}_{j-1}(\frac{k}{N})|^2 = \|\widetilde{V}_{j-1}\|^2 \quad (170e)$$

from an application of Equation (170c). Since the above holds for any $j \geq 2$, a proof by induction says that

$$\|\tilde{\mathbf{V}}_1\|^2 = \sum_{j=2}^{J_0} \|\tilde{\mathbf{W}}_j\|^2 + \|\tilde{\mathbf{V}}_{J_0}\|^2$$

for any $J_0 \geq 2$ also. The energy decomposition of Equation (169c) is now established if we can show that $\|\mathbf{X}\|^2 = \|\tilde{\mathbf{W}}_1\|^2 + \|\tilde{\mathbf{V}}_1\|^2$; in fact, Exercise [167b] states that this is true when $N \geq L$, but this restriction is removed in the following exercise.

▷ **Exercise [171a]:** Show that

$$\|\mathbf{X}\|^2 = \|\tilde{\mathbf{W}}_1\|^2 + \|\tilde{\mathbf{V}}_1\|^2$$

by using an argument that parallels the one leading to (170e).

We can use the energy decomposition of Equation (169c) to provide a MODWT-based analysis of the sample variance of $\mathbf{X}$ as follows. Since $\hat{\sigma}_X^2 = \frac{1}{N}\|\mathbf{X}\|^2 - \bar{X}^2$ (as indicated in Equation (48b)), it follows that

$$\hat{\sigma}_X^2 = \frac{1}{N} \sum_{j=1}^{J_0} \|\tilde{\mathbf{W}}_j\|^2 + \frac{1}{N} \|\tilde{\mathbf{V}}_{J_0}\|^2 - \bar{X}^2 \quad (171a)$$

(the above closely parallels the DWT-based ANOVA given in Equation (104c)). The following exercise shows that the last two terms above can be regarded as the sample variance of $\tilde{\mathbf{V}}_{J_0}$.

▷ **Exercise [171b]:** Show that the sample mean of $\tilde{\mathbf{V}}_{J_0}$ is equal to $\bar{X}$ and hence that the sample variance of $\tilde{\mathbf{V}}_{J_0}$ is given by $\frac{1}{N}\|\tilde{\mathbf{V}}_{J_0}\|^2 - \bar{X}^2$.

In preparation for establishing the MODWT-based MRA of Equation (169d), we first need to define $\tilde{\mathcal{D}}_j$ and $\tilde{\mathcal{S}}_j$. To do so, first note that, in matrix notation, we can express the transforms from $\mathbf{X}$ to $\tilde{\mathbf{W}}_j$ and from $\mathbf{X}$ to $\tilde{\mathbf{V}}_j$ as

$$\tilde{\mathbf{W}}_j = \tilde{\mathcal{W}}_j \mathbf{X} \text{ and } \tilde{\mathbf{V}}_j = \tilde{\mathcal{V}}_j \mathbf{X}, \quad (171b)$$

where each row of the $N \times N$ matrix $\tilde{\mathcal{W}}_j$ has values dictated by $\{\tilde{h}_{j,l}^\circ\}$, while $\tilde{\mathcal{V}}_j$ has values dictated by $\{\tilde{g}_{j,l}^\circ\}$; i.e., in view of Equation (170a), we must have

$$\tilde{\mathcal{W}}_j \equiv \begin{bmatrix} \tilde{h}_{j,0}^\circ & \tilde{h}_{j,N-1}^\circ & \tilde{h}_{j,N-2}^\circ & \tilde{h}_{j,N-3}^\circ & \cdots & \tilde{h}_{j,3}^\circ & \tilde{h}_{j,2}^\circ & \tilde{h}_{j,1}^\circ \\ \tilde{h}_{j,1}^\circ & \tilde{h}_{j,0}^\circ & \tilde{h}_{j,N-1}^\circ & \tilde{h}_{j,N-2}^\circ & \cdots & \tilde{h}_{j,4}^\circ & \tilde{h}_{j,3}^\circ & \tilde{h}_{j,2}^\circ \\ \tilde{h}_{j,2}^\circ & \tilde{h}_{j,1}^\circ & \tilde{h}_{j,0}^\circ & \tilde{h}_{j,N-1}^\circ & \cdots & \tilde{h}_{j,5}^\circ & \tilde{h}_{j,4}^\circ & \tilde{h}_{j,3}^\circ \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ \tilde{h}_{j,N-2}^\circ & \tilde{h}_{j,N-3}^\circ & \tilde{h}_{j,N-4}^\circ & \tilde{h}_{j,N-5}^\circ & \cdots & \tilde{h}_{j,1}^\circ & \tilde{h}_{j,0}^\circ & \tilde{h}_{j,N-1}^\circ \\ \tilde{h}_{j,N-1}^\circ & \tilde{h}_{j,N-2}^\circ & \tilde{h}_{j,N-3}^\circ & \tilde{h}_{j,N-4}^\circ & \cdots & \tilde{h}_{j,2}^\circ & \tilde{h}_{j,1}^\circ & \tilde{h}_{j,0}^\circ \end{bmatrix}, \quad (171c)$$

while $\tilde{\mathcal{V}}_j$ is expressed as above with each $\tilde{h}_{j,l}^\circ$ replaced by $\tilde{g}_{j,l}^\circ$. In analogy to Equation (95c), the desired definitions are

$$\tilde{\mathcal{D}}_j \equiv \tilde{\mathcal{W}}_j^T \tilde{\mathcal{W}}_j \text{ and } \tilde{\mathcal{S}}_j \equiv \tilde{\mathcal{V}}_j^T \tilde{\mathcal{V}}_j.$$

With these definitions, we can now prove the MRA of Equation (169d) as follows.

In view of what the rows of $\tilde{W}_j^T$ and $\tilde{V}_j^T$ look like (cf. Equation (171c)), we can express the elements of $\tilde{\mathcal{D}}_j$ and $\tilde{\mathcal{S}}_j$ as

$$\tilde{\mathcal{D}}_{j,t} = \sum_{l=0}^{N-1} \tilde{h}_{j,t}^\circ \tilde{W}_{j,t+l \bmod N} \quad \text{and} \quad \tilde{\mathcal{S}}_{j,t} = \sum_{l=0}^{N-1} \tilde{g}_{j,t}^\circ \tilde{V}_{j,t+l \bmod N},$$

$t = 0, 1, \dots, N-1$. Using an argument analogous to the one in Section 5.3 establishing that the elements of $\tilde{\mathcal{D}}_1$ are the output from a zero phase filter, we know that $\{\tilde{\mathcal{D}}_{j,t}\}$ is the result of circularly filtering $\{\tilde{W}_{j,t}\}$ with a filter whose DFT is $\{\tilde{H}_j^*(\frac{k}{N})\}$; likewise, $\{\tilde{\mathcal{S}}_{j,t}\}$ is obtained by circularly filtering $\{\tilde{V}_{j,t}\}$ with the filter $\{\tilde{h}_{j,t}^\circ\}$ whose DFT is $\{\tilde{H}_j(\frac{k}{N})\}$. Letting $\{\mathcal{X}_k\}$ represent the DFT of $\{X_t\}$ again, it follows that

$$\{\tilde{\mathcal{D}}_{j,t}\} \longleftrightarrow \{\tilde{H}_j^*(\frac{k}{N}) \tilde{H}_j(\frac{k}{N}) \mathcal{X}_k\} = \{|\tilde{H}_j(\frac{k}{N})|^2 \mathcal{X}_k\}; \quad (172a)$$

likewise, we have

$$\{\tilde{\mathcal{S}}_{j,t}\} \longleftrightarrow \{|\tilde{G}_j(\frac{k}{N})|^2 \mathcal{X}_k\}. \quad (172b)$$

Since $\{a_t\} \longleftrightarrow \{A_k\}$ and $\{b_t\} \longleftrightarrow \{B_k\}$ implies $\{a_t + b_t\} \longleftrightarrow \{A_k + B_k\}$ (an easy exercise!), we have

$$\{\tilde{\mathcal{D}}_{j,t} + \tilde{\mathcal{S}}_{j,t}\} \longleftrightarrow \{(|\tilde{H}_j(\frac{k}{N})|^2 + |\tilde{G}_j(\frac{k}{N})|^2) \mathcal{X}_k\}.$$

Equation (170d) says that $|\tilde{H}_j(\frac{k}{N})|^2 + |\tilde{G}_j(\frac{k}{N})|^2 = |\tilde{G}_{j-1}(\frac{k}{N})|^2$ when $j \geq 2$, so the above becomes

$$\{\tilde{\mathcal{D}}_{j,t} + \tilde{\mathcal{S}}_{j,t}\} \longleftrightarrow \{|\tilde{G}_{j-1}(\frac{k}{N})|^2 \mathcal{X}_k\};$$

however, Equation (172b) says that $\{\tilde{\mathcal{S}}_{j-1,t}\} \longleftrightarrow \{|\tilde{G}_{j-1}(\frac{k}{N})|^2 \mathcal{X}_k\}$ also, so we must have $\tilde{\mathcal{S}}_{j-1,t} = \tilde{\mathcal{D}}_{j,t} + \tilde{\mathcal{S}}_{j,t}$ for all t , establishing that

$$\tilde{\mathcal{S}}_{j-1} = \tilde{\mathcal{D}}_j + \tilde{\mathcal{S}}_j \quad (172c)$$

for all $j \geq 2$. An easy induction argument says that

$$\tilde{\mathcal{S}}_1 = \sum_{j=2}^{J_0} \tilde{\mathcal{D}}_j + \tilde{\mathcal{S}}_{J_0}$$

for any $J_0 \geq 2$. The MRA of Equation (169d) is now established if we can show that $\mathbf{X} = \tilde{\mathcal{S}}_1 + \tilde{\mathcal{D}}_1$; although Exercise [167b] says that this holds when $N \geq L$, we can remove this restriction in the following exercise.

▷ **Exercise [172]:** Show that

$$\mathbf{X} = \tilde{\mathcal{S}}_1 + \tilde{\mathcal{D}}_1$$

by using an argument that parallels the one leading to Equation (172c).

Key Facts and Definitions in Section 5.4

For any sample size N , the MODWT wavelet and scaling coefficients are defined, respectively, by

$$\tilde{W}_{j,t} \equiv \sum_{l=0}^{L_j-1} \tilde{h}_{j,l} X_{t-l \bmod N} \quad \text{and} \quad \tilde{V}_{j,t} \equiv \sum_{l=0}^{L_j-1} \tilde{g}_{j,l} X_{t-l \bmod N},$$

$t = 0, \dots, N-1$, where $\{\tilde{h}_{j,l} : l = 0, \dots, L_j-1\}$ and $\{\tilde{g}_{j,l} : l = 0, \dots, L_j-1\}$ are, respectively, the j th level MODWT wavelet and scaling filters, which are defined in terms of the j th level wavelet and scaling filters $\{h_{j,l}\}$ and $\{g_{j,l}\}$ via $\tilde{h}_{j,l} \equiv h_{j,l}/2^{j/2}$ and $\tilde{g}_{j,l} \equiv g_{j,l}/2^{j/2}$ (here $L_j \equiv (2^j - 1)(L - 1) + 1$ as before). In matrix notation, we can express the above as

$$\tilde{\mathbf{W}}_j = \tilde{\mathbf{W}}_j \mathbf{X} \quad \text{and} \quad \tilde{\mathbf{V}}_j = \tilde{\mathbf{V}}_j \mathbf{X},$$

where the rows of $\tilde{\mathbf{W}}_j$ contain circularly shifted versions of $\{\tilde{h}_{j,l}\}$ (i.e., $\{\tilde{h}_{j,l}\}$ periodized to length N), while the rows of $\tilde{\mathbf{V}}_j$ contain circularly shifted versions of $\{\tilde{g}_{j,l}\}$. The transfer functions for the j th MODWT wavelet and scaling filters are given by, respectively,

$$\tilde{H}(f) \equiv \tilde{H}(2^{j-1}f) \prod_{l=0}^{j-2} \tilde{G}(2^l f) \quad \text{and} \quad \tilde{G}_j(f) \equiv \prod_{l=0}^{j-1} \tilde{G}(2^l f),$$

where, $\tilde{H}(\cdot)$ and $\tilde{G}(\cdot)$ are the transfer functions for $\{\tilde{h}_l\} \equiv \{h_l/\sqrt{2}\}$ and $\{\tilde{g}_l\} \equiv \{g_l/\sqrt{2}\}$. With the definition $\tilde{\mathbf{V}}_0 = \mathbf{X}$, we have

$$\|\tilde{\mathbf{W}}_j\|^2 + \|\tilde{\mathbf{V}}_j\|^2 = \|\tilde{\mathbf{V}}_{j-1}\|^2$$

for $j = 1, 2, \dots$, leading to the MODWT energy decomposition

$$\|\mathbf{X}\|^2 = \sum_{j=1}^{J_0} \|\tilde{\mathbf{W}}_j\|^2 + \|\tilde{\mathbf{V}}_{J_0}\|^2$$

for any $J_0 \geq 1$. The j th level MODWT detail and smooth are defined by, respectively,

$$\tilde{\mathcal{D}}_j \equiv \tilde{\mathbf{W}}_j^T \tilde{\mathbf{W}}_j \quad \text{and} \quad \tilde{\mathcal{S}}_j \equiv \tilde{\mathbf{V}}_j^T \tilde{\mathbf{V}}_j,$$

in terms of which we can express the MODWT additive decomposition

$$\mathbf{X} = \sum_{j=1}^{J_0} \tilde{\mathcal{D}}_j + \tilde{\mathcal{S}}_{J_0}$$

for any $J_0 \geq 1$.

Comments and Extensions to Section 5.4

[1] If the sample size N is a power of two, the DWT of the time series $\mathbf{X}$ can be extracted from the MODWT because of the relationship $W_{j,t} = 2^{j/2} \tilde{W}_{j,2^j(t+1)-1}$. It is also true that, for any m , the DWT of the circularly shifted series $\mathcal{T}^m \mathbf{X}$ can be extracted from the MODWT. Liang and Parks (1996) used this fact in defining an orthonormal ‘translation invariant’ DWT (similar ideas are considered by Del Marco and Weiss, 1997). They considered the DWTs of $\mathcal{T}^m \mathbf{X}$, $m = 0, \dots, N-1$, and picked the one that maximizes a certain energy concentration measure. This procedure yields the shifted DWT that is best matched to $\mathbf{X}$ in the sense of concentrating a large portion of the DWT energy in as few coefficients as possible. This transform is indeed insensitive to circular shifts because it takes into account all possible circular shifts of $\mathbf{X}$ and hence will yield the same result if we were to use any $\mathcal{T}^m \mathbf{X}$ as our original time series in place of $\mathbf{X}$.

Since the DWT requires $O(N)$ multiplications, a brute force approach to computing all N of the DWTs needed to find the translation invariant DWT would take $O(N^2)$ multiplications, but Liang and Parks noted that the MODWT can compute all the required coefficients in just $O(N \log_2 N)$ multiplications. The fundamental reason for this reduction is that the DWTs for particular collections of $\mathcal{T}^m \mathbf{X}$ ’s have the same coefficients at particular scales. For example, $\mathcal{T}^m \mathbf{X}$ for $m = 0, 2, \dots, N-2$ all have the same $j = 1$ coefficients, as do $\mathcal{T}^m \mathbf{X}$ for $m = 1, 3, \dots, N-1$; $\mathcal{T}^m \mathbf{X}$ for $m = 0, 4, \dots, N-4$ all share the same $j = 2$ coefficients, as do $\mathcal{T}^m \mathbf{X}$ for either $m = 1, 5, \dots, N-3$, $m = 2, 6, \dots, N-2$ or $m = 3, 7, \dots, N-1$; and so forth.

[2] In Section 4.6, we argued that it is reasonable to associate the DWT scaling coefficients $\{V_{j,t}\}$ with scale $\lambda_j \equiv 2^j$ because the autocorrelation width for the scaling filter $\{g_{j,l}\}$ is always equal to λ_j :

$$\text{width}_a \{g_{j,l}\} \equiv \frac{(\sum_{l=-\infty}^{\infty} g_{j,l})^2}{\sum_{l=-\infty}^{\infty} g_{j,l}^2} = \lambda_j$$

(this is Equation (103)). Since $\tilde{g}_{j,l} \equiv g_{j,l}/2^{j/2}$, it is easy to see that $\text{width}_a \{\tilde{g}_{j,l}\} = \lambda_j$ also because the factor $1/2^{j/2}$ cancels out in the ratio above. We can thus claim that the MODWT scaling coefficients $\{\tilde{V}_{j,t}\}$ are associated with averages over scale λ_j and, using the same line of reasoning as in the DWT case, that the MODWT wavelet coefficients $\{\tilde{W}_{j,t}\}$ are associated with changes in averages over scale $\tau_j \equiv 2^{j-1}$.

5.5 Pyramid Algorithm for the MODWT

Here we describe an efficient algorithm for computing the j th level MODWT wavelet and scaling coefficients $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ based upon the scaling coefficients $\tilde{\mathbf{V}}_{j-1}$ of level $j-1$ (pseudo-code for this algorithm is given in item [1] of the Comments and Extensions). The key to this algorithm is to note the relationship between the filters used to compute the coefficients of levels $j-1$ and j . Equation (170a) states that the elements of $\tilde{\mathbf{W}}_j$, $\tilde{\mathbf{V}}_j$ and $\tilde{\mathbf{V}}_{j-1}$ are obtained by circularly filtering $\{X_t\}$ with, respectively, the periodized filters $\{\tilde{h}_{j,l}\}$, $\{\tilde{g}_{j,l}^\circ\}$ and $\{\tilde{g}_{j-1,l}^\circ\}$. The DFTs of these filters are, respectively, $\{\tilde{H}_j(\frac{k}{N})\}$, $\{\tilde{G}_j(\frac{k}{N})\}$ and $\{\tilde{G}_{j-1}(\frac{k}{N})\}$. An application of Equation (169b) tells us that the first two of these DFTs can be expressed in terms of the third DFT and the transfer functions $\tilde{H}(\cdot)$ and $\tilde{G}(\cdot)$ for the basic wavelet and scaling filters $\{\tilde{h}_l\}$ and $\{\tilde{g}_l\}$ in the following way:

$$\tilde{H}_j(\frac{k}{N}) = \tilde{G}_{j-1}(\frac{k}{N}) \tilde{H}(2^{j-1} \frac{k}{N}) \quad \text{and} \quad \tilde{G}_j(\frac{k}{N}) = \tilde{G}_{j-1}(\frac{k}{N}) \tilde{G}(2^{j-1} \frac{k}{N}).$$

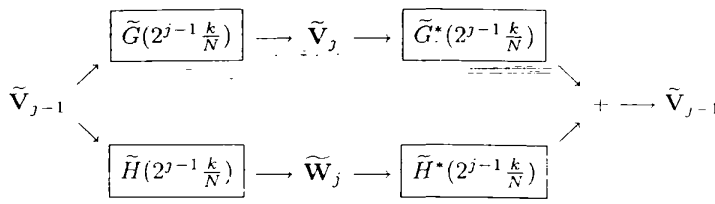


Figure 175. Flow diagram illustrating analysis of $\tilde{\mathbf{V}}_{j-1}$ into $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$, followed by the synthesis of $\tilde{\mathbf{V}}_{j-1}$ from $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$.

Now the right-hand sides of the above equations describe filter cascades consisting of two circular filters. The first filter in both cases is the filter that – when applied to $\mathbf{X}$ – yields $\tilde{\mathbf{V}}_{j-1}$ as its output. If we were to take this output and feed it into the second circular filters, namely, those with DFTs $\{\tilde{H}(2^{j-1} \frac{k}{N})\}$ and $\{\tilde{G}(2^{j-1} \frac{k}{N})\}$, we would obtain $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ as outputs. Thus, we can obtain $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ by filtering $\tilde{\mathbf{V}}_{j-1}$ with the circular filters described by $\{\tilde{H}(2^{j-1} \frac{k}{N})\}$ and $\{\tilde{G}(2^{j-1} \frac{k}{N})\}$.

To see precisely what form these two filters take, recall the result of Exercise [91]: if the filter $\{\tilde{h}_l : l = 0, \dots, L-1\}$ of width L has transfer function $\tilde{H}(\cdot)$, then the filter of width $2^{j-1}(L-1) + 1$ with impulse response sequence

$$\tilde{h}_0, \underbrace{0, \dots, 0}_{2^{j-1}-1 \text{ zeros}}, \tilde{h}_1, \underbrace{0, \dots, 0}_{2^{j-1}-1 \text{ zeros}}, \dots, \tilde{h}_{L-2}, \underbrace{0, \dots, 0}_{2^{j-1}-1 \text{ zeros}}, \tilde{h}_{L-1} \quad (175a)$$

has a transfer function defined by $\tilde{H}(2^{j-1}f)$. Thus we can obtain the elements $\{\tilde{W}_{j,t}\}$ from $\{\tilde{V}_{j-1,t}\}$ using the formula

$$\tilde{W}_{j,t} = \sum_{l=0}^{L-1} \tilde{h}_l \tilde{V}_{j-1,t-2^{j-1}l \bmod N}, \quad t = 0, 1, \dots, N-1; \quad (175b)$$

using a similar argument, we also have

$$\tilde{V}_{j,t} = \sum_{l=0}^{L-1} \tilde{g}_l \tilde{V}_{j-1,t-2^{j-1}l \bmod N}, \quad t = 0, 1, \dots, N-1. \quad (175c)$$

These two equations constitute the MODWT pyramid algorithm. Note that, if we define $\tilde{V}_{0,t} = X_t$, the above two equations yield the first level MODWT wavelet and scaling coefficients $\tilde{\mathbf{W}}_1$ and $\tilde{\mathbf{V}}_1$ (see Equation (163e)).

Since it is possible to obtain $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ by filtering $\tilde{\mathbf{V}}_{j-1}$, it is not surprising that we can reconstruct $\tilde{\mathbf{V}}_{j-1}$ from $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$, as the following exercise indicates.

▷ **Exercise [175]:** Show that, by using filters whose DFTs are given by $\{\tilde{H}^*(2^{j-1} \frac{k}{N})\}$ and $\{\tilde{G}^*(2^{j-1} \frac{k}{N})\}$, the inverse MODWT can be computed via an inverse pyramid algorithm described by the following equation:

$$\tilde{V}_{j-1,t} = \sum_{l=0}^{L-1} \tilde{h}_l \tilde{W}_{j,t+2^{j-1}l \bmod N} + \sum_{l=0}^{L-1} \tilde{g}_l \tilde{V}_{j,t+2^{j-1}l \bmod N} \quad (175d)$$

for $t = 0, 1, \dots, N-1$.

Figure 175 depicts a flow diagram for the MODWT pyramid algorithm and its inverse.

We can describe the transforms from $\tilde{\mathbf{V}}_{j-1}$ to $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ using the $N \times N$ matrices $\tilde{\mathcal{B}}_j$ and $\tilde{\mathcal{A}}_j$:

$$\tilde{\mathbf{W}}_j = \tilde{\mathcal{B}}_j \tilde{\mathbf{V}}_{j-1} \text{ and } \tilde{\mathbf{V}}_j = \tilde{\mathcal{A}}_j \tilde{\mathbf{V}}_{j-1}.$$

The contents of these matrices are apparent from a study of Equations (175b) and (175c). The rows of $\tilde{\mathcal{B}}_j$ consist of the upsampled wavelet filter of Equation (175a) periodized to length N , with each row differing from its neighbors by circular shifts of one unit either forward or backward; likewise, the rows of $\tilde{\mathcal{A}}_j$ contain values dictated by the upsampled scaling filter. For example, for $j = 2$ and $N = 12$ with $L = 4$, we have

$$\tilde{\mathcal{B}}_2 \equiv \begin{bmatrix} \tilde{h}_0 & 0 & 0 & 0 & 0 & 0 & \tilde{h}_3 & 0 & \tilde{h}_2 & 0 & \tilde{h}_1 & 0 \\ 0 & \tilde{h}_0 & 0 & 0 & 0 & 0 & 0 & \tilde{h}_3 & 0 & \tilde{h}_2 & 0 & \tilde{h}_1 \\ \tilde{h}_1 & 0 & \tilde{h}_0 & 0 & 0 & 0 & 0 & \tilde{h}_3 & 0 & \tilde{h}_2 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \tilde{h}_3 & 0 & \tilde{h}_2 & 0 & \tilde{h}_1 & 0 & \tilde{h}_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \tilde{h}_3 & 0 & \tilde{h}_2 & 0 & \tilde{h}_1 & 0 & \tilde{h}_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \tilde{h}_3 & 0 & \tilde{h}_2 & 0 & \tilde{h}_1 & 0 & \tilde{h}_0 \end{bmatrix}, \quad (176)$$

while $\tilde{\mathcal{A}}_2$ is obtained by replacing the $\tilde{h}_l$'s in the above with $\tilde{g}_l$'s. Note that this description is consistent with our formulation of $\tilde{\mathcal{B}}_1$ and $\tilde{\mathcal{A}}_1$ in Section 5.3. With $\tilde{\mathcal{B}}_j$ and $\tilde{\mathcal{A}}_j$, we can express the MODWT transform from $\tilde{\mathbf{V}}_{j-1}$ to $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ in a manner analogous to the DWT transform. Thus we have

$$\begin{bmatrix} \tilde{\mathbf{W}}_j \\ \tilde{\mathbf{V}}_j \end{bmatrix} = \tilde{\mathcal{P}}_j \tilde{\mathbf{V}}_{j-1} = \begin{bmatrix} \tilde{\mathcal{B}}_j \\ \tilde{\mathcal{A}}_j \end{bmatrix} \tilde{\mathbf{V}}_{j-1} \text{ with } \tilde{\mathcal{P}}_j \equiv \begin{bmatrix} \tilde{\mathcal{B}}_j \\ \tilde{\mathcal{A}}_j \end{bmatrix},$$

which is seen to be identical to Equation (94b) with the placement of tildes over all the components, indicating use of the MODWT filters. Furthermore, it is apparent from Equation (175d) that the synthesis of $\tilde{\mathbf{V}}_{j-1}$ from $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ can also be expressed in terms of $\tilde{\mathcal{B}}_j$ and $\tilde{\mathcal{A}}_j$:

$$\tilde{\mathbf{V}}_{j-1} = \tilde{\mathcal{B}}_j^T \tilde{\mathbf{W}}_j + \tilde{\mathcal{A}}_j^T \tilde{\mathbf{V}}_j.$$

If we recall that $\tilde{\mathbf{V}}_0 \equiv \mathbf{X}$, then recursive application of the above out to stage J_0 yields

$$\begin{aligned} \mathbf{X} &= \tilde{\mathcal{B}}_1^T \tilde{\mathbf{W}}_1 + \tilde{\mathcal{A}}_1^T \tilde{\mathcal{B}}_2^T \tilde{\mathbf{W}}_2 + \tilde{\mathcal{A}}_1^T \tilde{\mathcal{A}}_2^T \tilde{\mathcal{B}}_3^T \tilde{\mathbf{W}}_3 + \cdots + \tilde{\mathcal{A}}_1^T \cdots \tilde{\mathcal{A}}_{J_0-1}^T \tilde{\mathcal{B}}_{J_0}^T \tilde{\mathbf{W}}_{J_0} \\ &\quad + \tilde{\mathcal{A}}_1^T \cdots \tilde{\mathcal{A}}_{J_0-1}^T \tilde{\mathcal{A}}_{J_0}^T \tilde{\mathbf{V}}_{J_0}. \end{aligned}$$

We can use the above to identify expressions for the j th MODWT detail $\tilde{\mathcal{D}}_j$ and the J_0 th MODWT smooth $\tilde{\mathcal{S}}_{J_0}$, namely,

$$\tilde{\mathcal{D}}_j = \tilde{\mathcal{A}}_1^T \cdots \tilde{\mathcal{A}}_{j-1}^T \tilde{\mathcal{B}}_j^T \tilde{\mathbf{W}}_j \text{ and } \tilde{\mathcal{S}}_{J_0} = \tilde{\mathcal{A}}_1^T \cdots \tilde{\mathcal{A}}_{J_0-1}^T \tilde{\mathcal{A}}_{J_0}^T \tilde{\mathbf{V}}_{J_0}.$$

Comparison of the above with the definitions $\tilde{\mathcal{D}}_j \equiv \tilde{\mathcal{W}}_j^T \tilde{\mathbf{W}}_j$ and $\tilde{\mathcal{S}}_{J_0} \equiv \tilde{\mathcal{V}}_{J_0}^T \tilde{\mathbf{V}}_{J_0}$ yields expressions for $\tilde{\mathcal{W}}_j$ and $\tilde{\mathcal{V}}_j$ in terms of the $\mathcal{B}_j$ and $\mathcal{A}_j$ matrices:

$$\tilde{\mathcal{W}}_j = \tilde{\mathcal{B}}_j \tilde{\mathcal{A}}_{j-1} \cdots \tilde{\mathcal{A}}_1 \text{ and } \tilde{\mathcal{V}}_j = \tilde{\mathcal{A}}_j \tilde{\mathcal{A}}_{j-1} \cdots \tilde{\mathcal{A}}_1.$$

Note that Equation (171b) says that we can obtain $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$ by premultiplying $\mathbf{X}$ individually by the above $N \times N$ matrices.

Key Facts and Definitions in Section 5.5

Although by definition

$$\widetilde{W}_{j,t} \equiv \sum_{l=0}^{L_j-1} \tilde{h}_{j,l} X_{t-l \bmod N} \quad \text{and} \quad \widetilde{V}_{j,t} \equiv \sum_{l=0}^{L_j-1} \tilde{g}_{j,l} X_{t-l \bmod N},$$

$t = 0, \dots, N-1$, we can also obtain these j th level MODWT coefficients from the MODWT scaling coefficients of level $j-1$ via the recursions

$$\widetilde{W}_{j,t} = \sum_{l=0}^{L-1} \tilde{h}_l \widetilde{V}_{j-1,t-2^{j-1}l \bmod N} \quad \text{and} \quad \widetilde{V}_{j,t} = \sum_{l=0}^{L-1} \tilde{g}_l \widetilde{V}_{j-1,t-2^{j-1}l \bmod N}$$

(here we define $\widetilde{V}_{0,t} \equiv X_t$). We can describe the above in matrix notation as

$$\widetilde{\mathbf{W}}_j = \widetilde{\mathbf{B}}_j \widetilde{\mathbf{V}}_{j-1} \quad \text{and} \quad \widetilde{\mathbf{V}}_j = \widetilde{\mathbf{A}}_j \widetilde{\mathbf{V}}_{j-1},$$

where the rows of $\widetilde{\mathbf{B}}_j$ contain circularly shifted versions of $\{\tilde{h}_l\}$ after it has been upsampled to width $2^{j-1}(L-1)+1$ and then periodized to length N , with a similar construction for $\widetilde{\mathbf{A}}_j$ based upon $\{\tilde{g}_l\}$ (here upsampling consists of inserting $2^{j-1}-1$ zeros between each of the L values of the original filter). We can reconstruct $\{\widetilde{V}_{j-1,t}\}$ from $\{\widetilde{W}_{j,t}\}$ and $\{\widetilde{V}_{j,t}\}$ via

$$\widetilde{V}_{j-1,t} = \sum_{l=0}^{L-1} \tilde{h}_l \widetilde{W}_{j,t+2^{j-1}l \bmod N} + \sum_{l=0}^{L-1} \tilde{g}_l \widetilde{V}_{j,t+2^{j-1}l \bmod N},$$

which can be expressed in matrix notation as $\widetilde{\mathbf{V}}_{j-1} = \widetilde{\mathbf{B}}_j^T \widetilde{\mathbf{W}}_j + \widetilde{\mathbf{A}}_j^T \widetilde{\mathbf{V}}_j$. The above relationships allow us to express the MODWT details and smooths as

$$\widetilde{\mathcal{D}}_j = \widetilde{\mathcal{A}}_1^T \cdots \widetilde{\mathcal{A}}_{j-1}^T \widetilde{\mathbf{B}}_j^T \widetilde{\mathbf{W}}_j \quad \text{and} \quad \widetilde{\mathcal{S}}_{J_0} = \widetilde{\mathcal{A}}_1^T \cdots \widetilde{\mathcal{A}}_{J_0-1}^T \widetilde{\mathcal{A}}_{J_0}^T \widetilde{\mathbf{V}}_{J_0},$$

which in turn says that the matrices $\widetilde{\mathbf{W}}_j$ and $\widetilde{\mathbf{V}}_j$ can be expressed as

$$\widetilde{\mathbf{W}}_j = \widetilde{\mathbf{B}}_j \widetilde{\mathcal{A}}_{j-1} \cdots \widetilde{\mathcal{A}}_1 \quad \text{and} \quad \widetilde{\mathbf{V}}_j = \widetilde{\mathcal{A}}_j \widetilde{\mathcal{A}}_{j-1} \cdots \widetilde{\mathcal{A}}_1.$$

(these matrices yield $\widetilde{\mathbf{W}}_j = \widetilde{\mathcal{W}}_j \mathbf{X}$ and $\widetilde{\mathbf{V}}_j = \widetilde{\mathcal{V}}_j \mathbf{X}$).

Comments and Extensions to Section 5.5

[1] Here we give pseudo-code that can be used to compute the MODWT and its inverse. Define $\widetilde{V}_{0,t} \equiv X_t$ for $t = 0, \dots, N-1$, where N is any positive integer. Let $\{\tilde{h}_l : l = 0, \dots, L-1\}$ be a MODWT wavelet filter of even width L . Let $\{\tilde{g}_l\}$ be the corresponding MODWT scaling filter constructed from $\{\tilde{h}_l\}$ via Equation (163d). Denote the elements of $\widetilde{\mathbf{W}}_j$ and $\widetilde{\mathbf{V}}_{J_0}$ as $\widetilde{W}_{j,t}$ and $\widetilde{V}_{J_0,t}$, $t = 0, \dots, N-1$. Given $\widetilde{\mathbf{V}}_{j-1}$, the following pseudo-code computes $\widetilde{\mathbf{W}}_j$ and $\widetilde{\mathbf{V}}_j$ (this is based on Equations (175b) and (175c)):

For $t = 0, \dots, N - 1$, do the outer loop:

Set k to t .

Set $\tilde{W}_{j,t}$ to $\tilde{h}_0 \tilde{V}_{j-1,k}$.

Set $\tilde{V}_{j,t}$ to $\tilde{g}_0 \tilde{V}_{j-1,k}$.

For $n = 1, \dots, L - 1$, do the inner loop:

Decrement k by 2^{j-1} .

If $k < 0$, set k to $k \bmod N$.

Increment $\tilde{W}_{j,t}$ by $\tilde{h}_n \tilde{V}_{j-1,k}$.

Increment $\tilde{V}_{j,t}$ by $\tilde{g}_n \tilde{V}_{j-1,k}$.

End of inner loop.

End of outer loop.

Note that, if j is such that $2^{j-1} \leq N$ (as will usually be the case in practical applications), then the single ' $k \bmod N$ ' in the above code can be replaced by ' $k + N$ '.

By using the pseudo-code above first for $j = 1$ (i.e., starting with $\tilde{\mathbf{V}}_0 = \mathbf{X}$) and then for $j = 2, \dots, J_0$, we obtain the component vectors of the MODWT, namely, $\tilde{\mathbf{W}}_1, \dots, \tilde{\mathbf{W}}_{J_0}$ and $\tilde{\mathbf{V}}_{J_0}$, along with the intermediate vectors $\tilde{\mathbf{V}}_1, \dots, \tilde{\mathbf{V}}_{J_0-1}$.

The task of the inverse MODWT is to compute $\mathbf{X}$ given the vectors $\tilde{\mathbf{W}}_1, \dots, \tilde{\mathbf{W}}_{J_0}$ and $\tilde{\mathbf{V}}_{J_0}$. Given $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$, the following pseudo-code computes $\tilde{\mathbf{V}}_{j-1}$ (this is based on Equation (175d)):

For $t = 0, \dots, N - 1$, do the outer loop:

Set k to t .

Set $\tilde{V}_{j-1,t}$ to $\tilde{h}_0 \tilde{W}_{j,k} + \tilde{g}_0 \tilde{V}_{j,k}$.

For $n = 1, \dots, L - 1$, do the inner loop:

Increment k by 2^{j-1} .

If $k \geq N$, set k to $k \bmod N$.

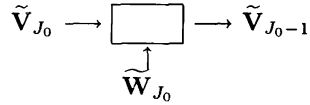
Increment $\tilde{V}_{j-1,t}$ by $\tilde{h}_n \tilde{W}_{j,k} + \tilde{g}_n \tilde{V}_{j,k}$.

End of the inner loop.

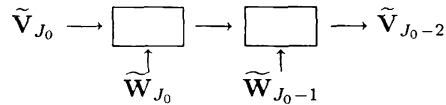
End of the outer loop.

Again, if j is such that $2^{j-1} \leq N$, then the single ' $k \bmod N$ ' in the above code can be replaced by ' $k - N$ '.

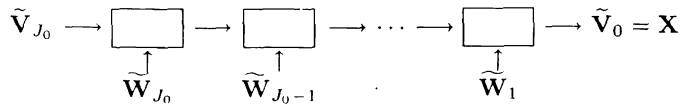
By using the pseudo-code first with $\tilde{\mathbf{W}}_{J_0}$ and $\tilde{\mathbf{V}}_{J_0}$, we obtain $\tilde{\mathbf{V}}_{J_0-1}$. We can depict this procedure as putting $\tilde{\mathbf{W}}_{J_0}$ and $\tilde{\mathbf{V}}_{J_0}$ into a box, out of which comes $\tilde{\mathbf{V}}_{J_0-1}$:



Using the pseudo-code next with $\tilde{\mathbf{W}}_{J_0-1}$ and $\tilde{\mathbf{V}}_{J_0-1}$ yields $\tilde{\mathbf{V}}_{J_0-2}$:

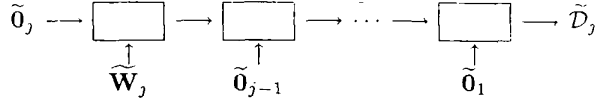


After $J_0 - 2$ more applications of the pseudo-code, we obtain $\tilde{\mathbf{V}}_0 = \mathbf{X}$:

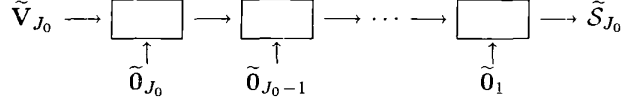


($\tilde{\mathbf{V}}_{J_0-3}, \dots, \tilde{\mathbf{V}}_1$ are produced as intermediate calculations).

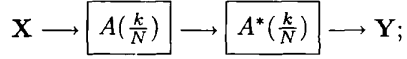
The j th detail $\tilde{\mathcal{D}}_j$ is obtained by taking the inverse MODWT of $\tilde{\mathbf{0}}_1, \dots, \tilde{\mathbf{0}}_{j-1}$, $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{0}}_j$, where each $\tilde{\mathbf{0}}_k$, $k = 1, \dots, j$, is a vector of N zeros. This detail can be computed by successively applying the inverse MODWT algorithm for $j, j-1, \dots, 1$ starting with $\tilde{\mathbf{W}}_j$ and with $\tilde{\mathbf{0}}_j$ substituted for $\tilde{\mathbf{V}}_j$ in the above pseudo-code. At the end of j iterations, we obtain the desired $\tilde{\mathcal{D}}_j$:



Likewise, the smooth $\tilde{\mathcal{S}}_{J_0}$ can be obtained by applying the inverse MODWT to $\tilde{\mathbf{0}}_1, \dots, \tilde{\mathbf{0}}_{J_0}$ and $\tilde{\mathbf{V}}_{J_0}$:



[2] The fact that the MODWT details and smooths are associated with zero phase filters is an example of a well-known ‘trick’ for obtaining a zero phase output based upon any arbitrary circular filter $\{a_l^o\} \longleftrightarrow \{A(\frac{k}{N})\}$. The trick (described in, for example, Hamming, 1989, p. 252) is to filter the time series $\mathbf{X}$ with $\{a_l^o\}$; reverse the output from this filter; and then filter the reversed output with $\{a_l^o\}$ again to obtain an output, which – when reversed again to obtain, say, $\mathbf{Y}$ – is associated with zero phase. Because filtering the reversed output, followed by reversal, is the same thing as cross correlation of the original output, this procedure is equivalent to the following flow diagram:



i.e., the equivalent circular filter for the two stage cascade has a frequency domain representation of $\{|A(\frac{k}{N})|^2\}$, which necessarily has zero phase.

5.6 MODWT Analysis of ‘Bump’ Time Series

Let us now revisit the example discussed in Section 5.1, in which we looked at a DWT analysis of a ‘bump’ time series $\mathbf{X}$ and its circular shift $\mathcal{T}^5 \mathbf{X}$. The bottom six rows of Figure 181 show the same quantities as in Figure 161, but now we use the MODWT instead of the DWT. Let us first consider the MODWT coefficients for $\mathbf{X}$ (first column, middle five rows). In contrast to the DWT, each of the MODWT coefficient vectors $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_4$ has the same number of elements as $\mathbf{X}$. As indicated by the labels in the first column, we have plotted each coefficient vector after advancing it (i.e., circularly shifting it to the left) by the amount indicated by the absolute value of the exponent for the shift operator $\mathcal{T}$ (see Equation (52b); the first and last values in, for example, $\mathcal{T}^{-53} \tilde{\mathbf{W}}_4$ are, respectively, $\tilde{W}_{4,53}$ and $\tilde{W}_{4,52}$). Because we are using the LA(8) wavelet filter in this example, these circular shifts advance the MODWT coefficient vectors so that they are approximately the outputs from a zero phase filter. The particular values of the advances for the wavelet and scaling coefficients are determined by the absolute values of, respectively, $\nu_j^{(H)}$ in Equation (114b) and $\nu_4^{(G)}$ in Equation (114a)

(in these equations, $\nu = -3$ for the LA(8) filter). The corresponding MODWT coefficient vectors for the shifted series $T^5\mathbf{X}$ are shown in the second column. A careful comparison of the first and second columns indicates that shifting $\mathbf{X}$ has the effect of shifting each of the MODWT coefficient vectors by a similar amount; i.e., if $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_{J_0}$ constitute the MODWT of $\mathbf{X}$, then $T^m\tilde{\mathbf{W}}_j$ and $T^m\tilde{\mathbf{V}}_{J_0}$ constitute the MODWT of $T^m\mathbf{X}$. In contrast to the DWT, circularly shifting $\mathbf{X}$ modifies the MODWT in an intuitively reasonable manner. Note also that the largest MODWT coefficients occur in $\tilde{\mathbf{W}}_3$, i.e., at changes in scale $\tau_3 = 4$. This seems reasonable, given that the bump is nonzero over just six values and has an effective width of 4.3 (using a measure provided by a finite sample size analog of Equation (103)).

Let us next compare the MRAs for $\mathbf{X}$ and $T^5\mathbf{X}$ – these are shown in, respectively, the third and fourth columns of Figure 181 (middle five rows). The MODWT details and smooth for the two series are again related in a simple and reasonable manner: if $\tilde{\mathcal{D}}_j$ and $\tilde{\mathcal{S}}_{J_0}$ form the MRA for $\mathbf{X}$, then $T^m\tilde{\mathcal{D}}_j$ and $T^m\tilde{\mathcal{S}}_{J_0}$ form the MRA for $T^m\mathbf{X}$. As is evident from Figure 181, this simple relationship does not hold for the DWT.

Finally, let us compare the MODWT coefficients and MRA for the bump series. As can be seen by looking at a particular row in the first and third columns of Figure 181, the vectors $T^{-|\nu_j^{(H)}|}\tilde{\mathbf{W}}_j$ and $\tilde{\mathcal{D}}_j$ are quite similar in appearance, as are $T^{-|\nu_4^{(G)}|}\tilde{\mathbf{V}}_4$ and $\tilde{\mathcal{S}}_4$. By contrast, the corresponding portion of Figure 161 shows that the DWT detail $\mathcal{D}_j$ can look quite different from $\mathbf{W}_j$ (partly because the former has 2^j more elements than the latter). To understand why $\tilde{\mathbf{W}}_j$ and $\tilde{\mathcal{D}}_j$ look so similar, recall that both vectors arise as outputs from circularly filtering $\mathbf{X}$. In the case of $\tilde{\mathbf{W}}_j$, the circular filter has a frequency domain representation of $\{H_j(\frac{k}{N})\}$. Because $T^{-|\nu_j^{(H)}|}\tilde{\mathbf{W}}_j$ is approximately the output from a zero phase filter, its associated circular filter is approximately $\{|H_j(\frac{k}{N})|\}$. On the other hand, we know from Equation (172a) that the circular filter yielding $\mathcal{D}_j$ is given by $\{|H_j(\frac{k}{N})|^2\}$. The circular filters yielding $T^{-|\nu_j^{(H)}|}\tilde{\mathbf{W}}_j$ and $\tilde{\mathcal{D}}_j$ thus have similar phase and gain properties (the fact that $H_j(\cdot)$ is approximately a band-pass filter means that $|H_j(\cdot)|^2$ must be also – it can be argued that the latter is a better approximation by certain measures). In the DWT case, we do not have such a tight correspondence, mainly because filtering is now accompanied by downsampling to form $\mathbf{W}_j$ and then upsampling to form $\mathcal{D}_j$.

Although the MODWT details and shifted wavelet coefficients are outputs from filters with similar phase and gain properties, the ways in which they can be put to good use are quite different. Let us demonstrate this using the bump series $\mathbf{X}$ and $T^5\mathbf{X}$. First, whereas Equation (169c) says that we can decompose the energy in $\mathbf{X}$ or $T^5\mathbf{X}$ using the MODWT coefficients, the same is not true for the details and smooth. To show this, we computed level $J_0 = 7$ MODWTs based again upon the LA(8) wavelet filter and used the wavelet coefficients to construct discrete wavelet empirical power spectra:

$$P_{\tilde{\mathbf{W}}}(\tau_j) \equiv \frac{1}{N} \|\tilde{\mathbf{W}}_j\|^2 = \frac{1}{N} \|T^5\tilde{\mathbf{W}}_j\|^2, \quad j = 1, \dots, 7.$$

Since there are $N = 128 = 2^7$ points in the bump series, we can use Exercise [5.9a] to claim that these spectra provide an analysis of the sample variance for $\mathbf{X}$ or $T^5\mathbf{X}$:

$$\hat{\sigma}_X^2 = \sum_{j=1}^7 P_{\tilde{\mathbf{W}}}(\tau_j)$$

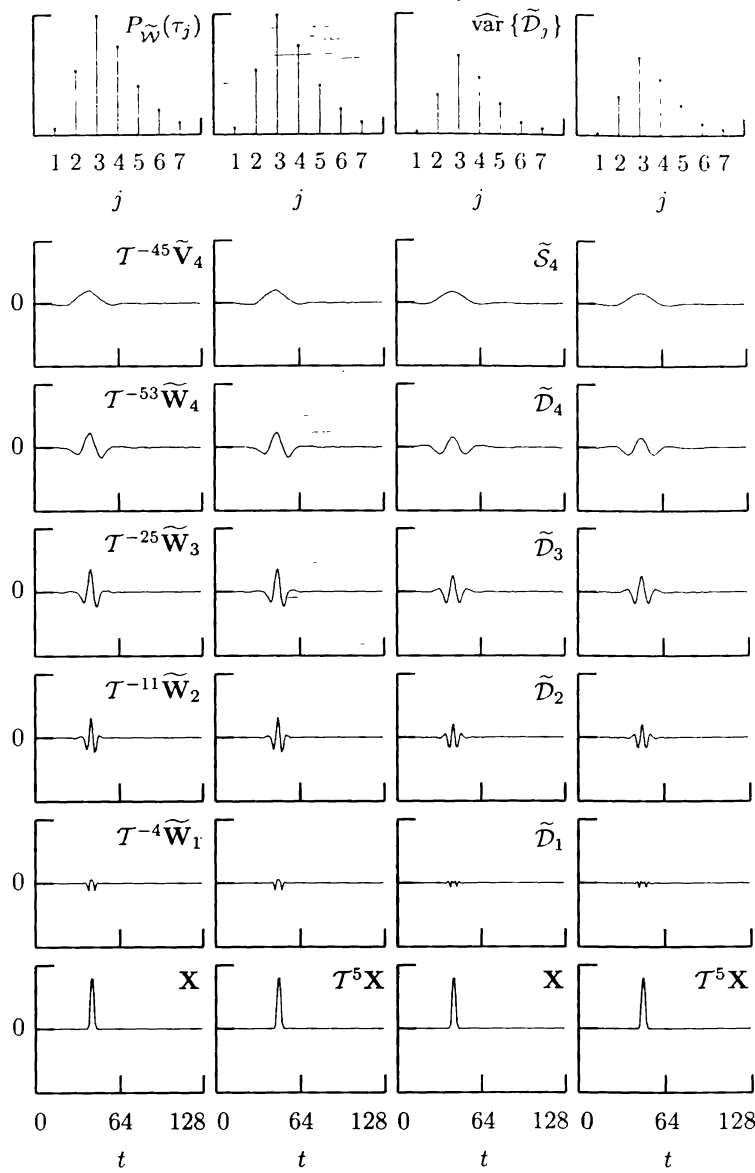


Figure 181. Bottom six rows as in Figure 161, with MODWT replacing DWT. The top row shows the discrete wavelet empirical power spectra (first two columns) and a corresponding quantity based upon level $j = 1, \dots, 7$ MODWT details (last two columns) – see text for details.

(the analysis of $\hat{\sigma}_X^2$ in terms of MODWT coefficients is studied further in Chapter 8). The spectra for X and T^5X are shown in the first two plots in the first row in Figure 181 and are seen to be identical – circularly shifting a series does not change its MODWT-based power spectrum (Equations (63a) and (63b) demonstrate that this is a true

in general for DWT-based spectra). To show that the corresponding details cannot be used in a similar manner, we have also computed

$$\widehat{\text{var}}\{\tilde{\mathcal{D}}_j\} \equiv \frac{1}{N}\|\tilde{\mathcal{D}}_j\|^2 = \frac{1}{N}\|\mathcal{T}^5\tilde{\mathcal{D}}_j\|^2, \quad j = 1, \dots, 7,$$

and plotted them for $\mathbf{X}$ and $\mathcal{T}^5\mathbf{X}$ as the last two plots in the first row. Exercise [5.7] claims that $\|\tilde{\mathcal{D}}_j\|^2 \leq \|\tilde{\mathbf{W}}_j\|^2$ always, so we must have $\widehat{\text{var}}\{\tilde{\mathcal{D}}_j\} \leq P_{\tilde{\mathbf{W}}}(\tau_j)$. Figure 181 indicates that this is true here. Any study of how the energy properties of $\mathbf{X}$ change over a scale of τ_j should thus be based on $\tilde{\mathbf{W}}_j$ rather than $\tilde{\mathcal{D}}_j$.

While MODWT details cannot be used to create a valid ANOVA, they are an integral part of the additive MRA decomposition. For the bump series and its shift, we have

$$\mathbf{X} = \sum_{j=1}^4 \tilde{\mathcal{D}}_j + \tilde{\mathcal{S}}_4 \quad \text{and} \quad \mathcal{T}^5\mathbf{X} = \sum_{j=1}^4 \mathcal{T}^5\tilde{\mathcal{D}}_j + \mathcal{T}^5\tilde{\mathcal{S}}_4,$$

so adding the series in the middle five rows of the last two columns of Figure 181 yields the time series in the bottom row. For this example,

$$\mathbf{X} \neq \sum_{j=1}^4 \mathcal{T}^{-|\nu_j^{(H)}|} \tilde{\mathbf{W}}_j + \mathcal{T}^{-|\nu_4^{(G)}|} \tilde{\mathbf{V}}_4,$$

so we cannot in general create an additive decomposition using the MODWT coefficients directly (Exercise [5.10], however, indicates an exception is the Haar DWT, whose coefficients *can* be used directly to provide an additive decomposition). The exact zero phase properties of the filters used to produce $\tilde{\mathcal{D}}_j$ and $\tilde{\mathcal{S}}_{J_0}$ are important for aligning events in the original series with those in an MRA, and the exact additivity provided by an MRA ensures that all the features in $\mathbf{X}$ are accounted for either in one of the details or in the smooth.

The interplay between use of the additive MRA and use of the MODWT coefficients for ANOVA is further illustrated in the sections that follow, where we consider MODWT-based wavelet analysis of actual time series.

5.7 Example: Electrocardiogram Data

Here we re-examine the ECG time series using the MODWT, mainly so that we can compare it with our previous analysis in Section 4.10 using the DWT. Figure 183 shows the MODWT of level $J_0 = 6$ based upon the LA(8) wavelet filter – the corresponding DWT analysis is shown in Figure 127. Prior to plotting, the coefficient vectors $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_6$ were circularly advanced by the factors $|\nu_j^{(H)}|$ of Equation (114b) and $|\nu_6^{(G)}|$ of Equation (114a), respectively, to align the coefficients properly as much as possible with the original time series (the amounts of the shifts are indicated by the exponent of each $\mathcal{T}$ recorded on the plot). The two thick vertical lines crossing a given $\mathcal{T}^{-|\nu_j^{(H)}|} \tilde{\mathbf{W}}_j$ or $\mathcal{T}^{-|\nu_6^{(G)}|} \tilde{\mathbf{V}}_6$ delineate regions at the beginning and end of the shifted vector that are affected at least to some degree by the assumption of circularity (we discuss how to compute the locations of these lines in Section 5.11). Judicious subsampling and rescaling of the MODWT coefficients yield the DWT coefficients. If we compare the coefficients for larger scales in Figure 127 with those in Figure 183, we see that the

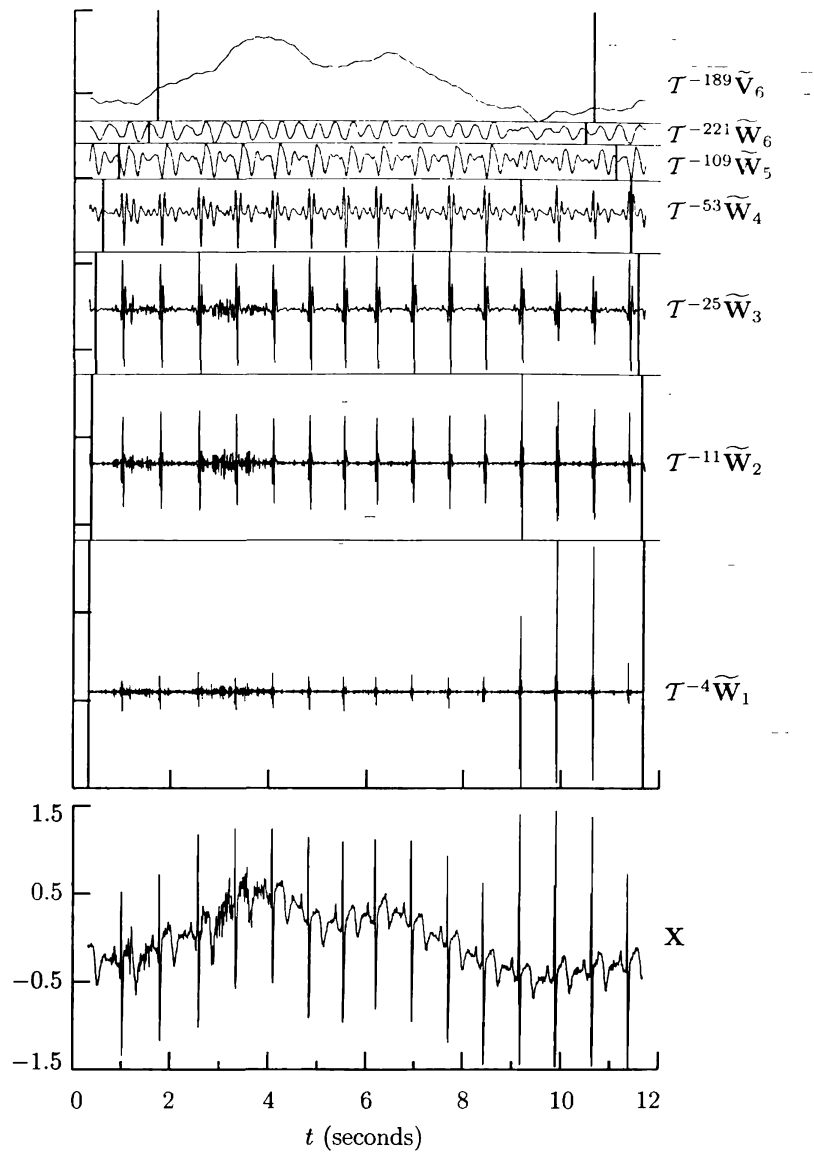


Figure 183. MODWT wavelet analysis of ECG time series using LA(8) wavelet. The above should be compared to Figure 127 for the corresponding DWT analysis (see text for details).

subsampling inherent in the DWT significantly hampers our ability to interpret its coefficients in terms of events in the original ECG series.

The MODWT multiresolution analysis is shown in Figure 184. As before, the thick vertical lines crossing a given $\tilde{D}_j$ or $\tilde{S}_6$ delineate regions at the beginning and end of the vector that are affected at least to some degree by the circularity assump-

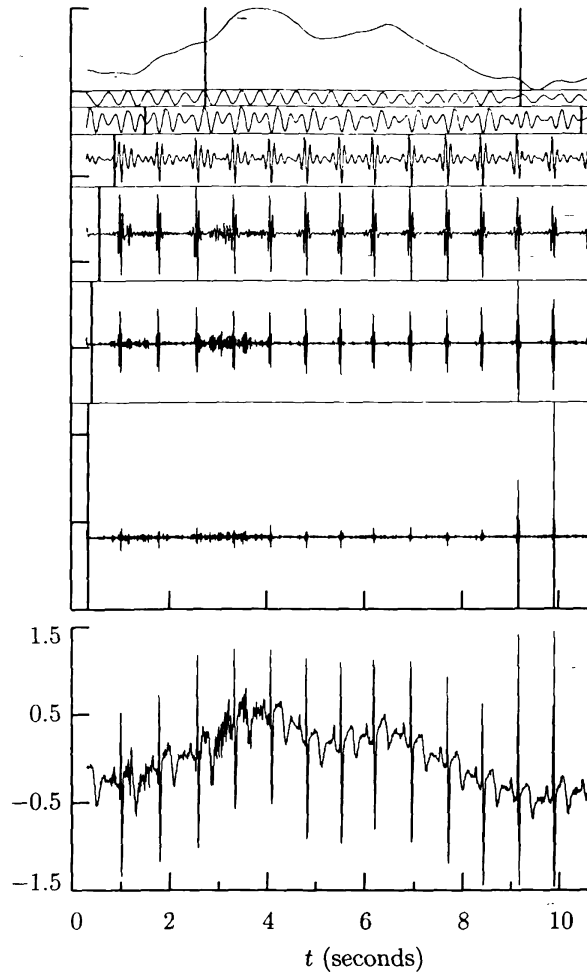


Figure 184. MODWT multiresolution analysis of ECG time series u (see text for details). Figure 133 shows the corresponding DWT analysis.

tion (see Section 5.11 for details). This MRA should be compared to the MRA shown in Figure 133. The overall appearances of the two MRAs are comparable, but note the difference between the DWT details $\mathcal{D}_3$ and the corresponding MODWT details $\tilde{\mathcal{D}}_3$, $\tilde{\mathcal{D}}_4$ and $\tilde{\mathcal{D}}_5$: in each case, the MODWT details are more consistent across time, indicating that the time variations seen in the DWT details are in part attributable to the particular alignment assumed by the DWT. The MODWT detail $\tilde{\mathcal{D}}_6$ does not tend to fade in and out like the DWT detail $\mathcal{D}_6$ (particularly at the beginning of the series), and the asymmetric appearance of the 'bumps' in $\mathcal{D}_6$ is also eliminated in the MODWT detail $\tilde{\mathcal{D}}_6$ due to the property associated with this detail.

Finally let us compare the MODWT wavelet and scaling coefficients in Figure 183 with the MODWT details and smooth in Figure 184. While the former preserves energy and the latter preserves additivity, the overall appearances of the two scale-by-scale decompositions are quite similar. The details and smooth are obtained by filtering the wavelet and scaling coefficients, so each $\tilde{\mathcal{D}}_j$ or $\tilde{\mathcal{S}}_6$ is necessarily smoother in appearance than the corresponding $\tilde{\mathcal{W}}_j$ or $\tilde{\mathcal{V}}_6$. This is particularly evident in a comparison of $\tilde{\mathcal{S}}_6$ with $\tilde{\mathcal{V}}_6$: the latter contains small ripples that are almost entirely absent in the former. While $\tilde{\mathcal{S}}_6$ is a quite useful estimate of the baseline drift here (and, likewise, $\tilde{\mathcal{S}}_{J_0}$ is a useful estimate of trend in other time series), any meaningful partitioning of the energy in the ECG series across scales and/or time must be done using the MODWT coefficients. Both types of analysis (the $\tilde{\mathcal{W}}_j$'s with $\tilde{\mathcal{V}}_{J_0}$ and the $\tilde{\mathcal{D}}_j$'s with $\tilde{\mathcal{S}}_{J_0}$) thus have an important role to play in the analysis of this and other time series.

5.8 Example: Subtidal Sea Level Fluctuations

In this section we illustrate MODWT-based MRA and ANOVA using a time series of subtidal sea levels for Crescent City, which is located on the open coast of Northern California. Inside its harbor, a permanent tide gauge is maintained by the National Ocean Service (NOS). The gauge measures water levels every 6 minutes within a stilling well, which suppresses high-frequency fluctuations due to wind waves and swell. Periodic leveling surveys ensure that the reference level of the gauge remains constant relative to the surrounding land. The time series considered here is based on a segment of the water level series from the beginning of 1980 to the end of 1991. This segment contains one gap two weeks in length in addition to a few relatively minor short gaps. The two week gap occurred in the summer of 1990 and was filled in using predicted tides plus local means, whereas interpolated values supplied by NOS were used to bridge the shorter gaps. The edited series was then subsampled every tenth value to produce hourly values and low-pass filtered using a Kaiser filter to remove the diurnal (daily) and semidiurnal (twice daily) tides. The 3 dB down (half-power) point of the squared gain function for the filter occurs at a frequency corresponding to a 1.82 day period, while frequencies corresponding to a 2.68 day period and longer are in the pass-band of the low-pass filter and are attenuated in power by less than 1%. Finally the filtered series was resampled every $\Delta t = 1/2$ day, yielding a time series $\{X_t\}$ measured in centimeters with $N = 8746$ values. This series reflects subtidal variations in the coastal sea level at Crescent City (i.e., variations after removal of the predictable tidal component) and is plotted at the bottom of Figure 186. (For more details about this application, see Percival and Mofjeld, 1997.)

Eight other series depicting a level $J_0 = 7$ LA(8) MODWT-based MRA are plotted above the Crescent City sea level data in Figure 186. The uppermost series is the MODWT smooth $\tilde{\mathcal{S}}_7$ (corresponding to averages over physical scales of $\lambda_7 \Delta t = 64$ days) while the seven series below it are the MODWT detail series $\tilde{\mathcal{D}}_1, \tilde{\mathcal{D}}_2, \dots, \tilde{\mathcal{D}}_7$ (corresponding to changes in physical scales of, from bottom to top, $1/2, 1, \dots, 32$ days). As usual, we have $\mathbf{X} = \sum_j \tilde{\mathcal{D}}_j + \tilde{\mathcal{S}}_7$. As explained in Section 5.11, the two solid vertical lines intersecting a given $\tilde{\mathcal{D}}_j$ or $\tilde{\mathcal{S}}_7$ delineate the regions of the MRA component affected to some degree (however small) by the assumption that $\{X_t\}$ is circularly defined for $t < 0$ and $t \geq N$; however, a circular extension of the series is reasonable here, given that we are looking at a phenomenon with an annual

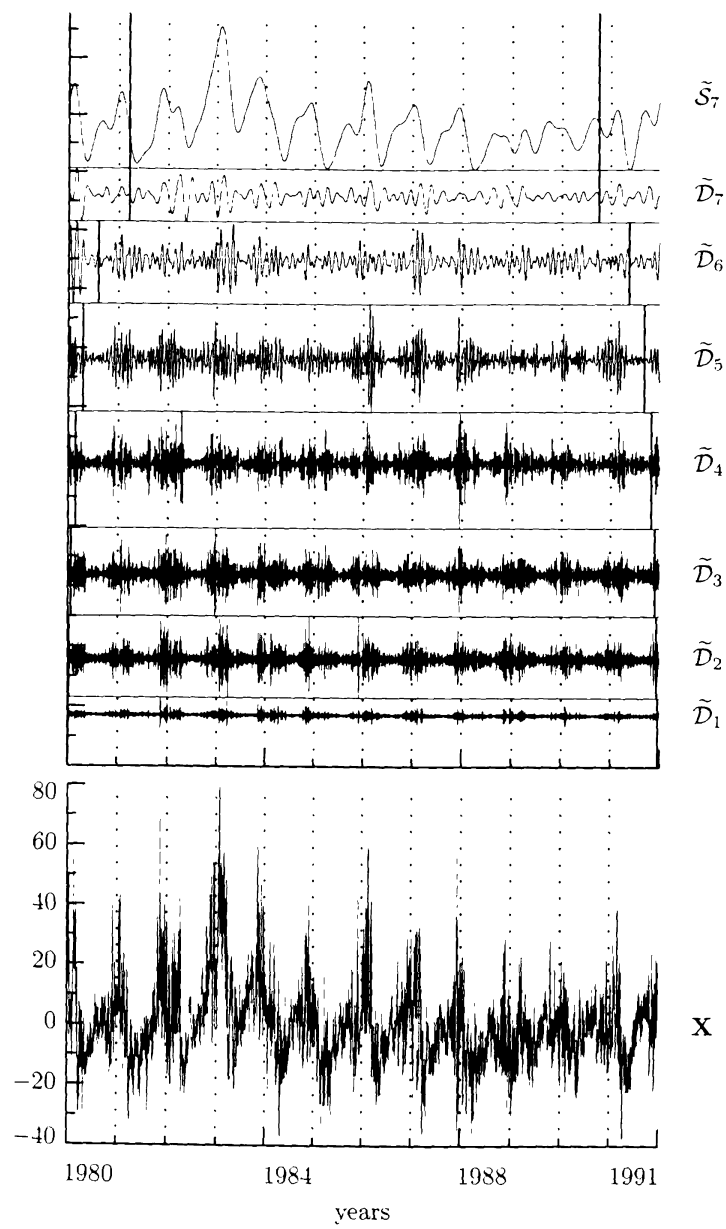


Figure 186. LA(8) MODWT multiresolution analysis for Crescent City subtidal variations (see text for details). This series is measured in centimeters.

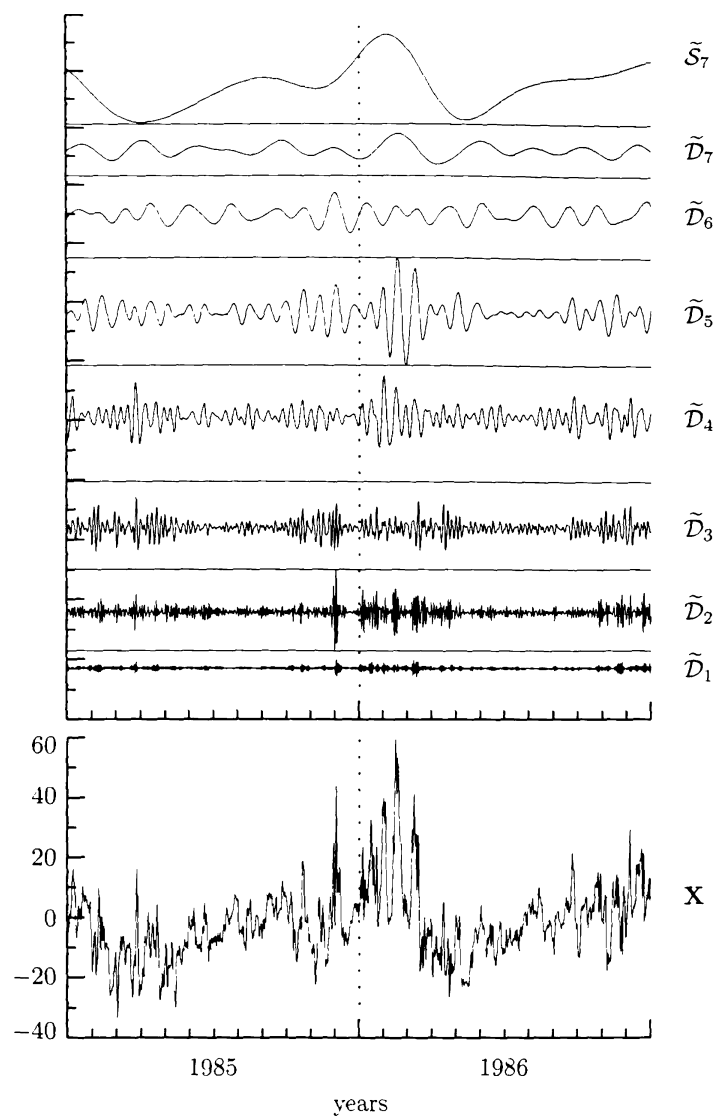


Figure 187. Expanded view of 1985 and 1986 portion of Figure 186.

component and that the amount of data we have is very close to 12 complete years.

Before we consider this MRA in more detail, however, let us comment upon why we decided to use the LA(8) wavelet filter and to set $J_0 = 7$. This particular filter was chosen because we eventually want to associate the MODWT coefficients themselves with actual times (had we just been interested in an MRA, a D(4) filter would have yielded nearly the same results, but detail series for small scales based on a Haar filter suffer from a form of ‘leakage’ in that they are noticeably tracking fluctuations from higher scales – see Section 8.9 for a discussion of leakage and how it is influenced by

the number of embedded differencing operations in various wavelet filters). The phase properties of the least asymmetric filters allow us to make associations with time, so we picked this class of filters and settled on the LA(8) filter in particular because the LA filters with $L = 10$ and 12 yielded quite similar analyses, but with necessarily greater concerns about the number of terms influenced by boundary conditions (see Section 5.11). The coiflet filters are also reasonable candidates, but, while the shortest member of this class – the C(6) coiflet – gave a very similar MRA, there was some evidence of leakage at the very shortest scale when its wavelet coefficients were used to carry out the subsequent ANOVA discussed in Section 8.7 (the next shortest coiflet is the C(12), which gives results like those of the LA(8) filter, but with the same concerns about boundary effects as for the LA(10) and LA(12) filters).

Our choice of level $J_0 = 7$ was based on the fact that the resulting smooth $\tilde{S}_7$ reflects averages over a physical scale of $\lambda_7 \Delta t = 2^7 \cdot \frac{1}{2} = 64$ days and hence contains the intra-annual variations in the subtidal series (i.e., variations between years). If we made $J_0 > 7$, we would break up the annual variations into pieces that are of not much interest to oceanographers as separate entities. On the other hand, if we set $J_0 < 7$, we would add some scales to $\tilde{S}_{J_0}$ that (as we shall see subsequently) have a seasonally dependent variability and hence are better studied separately.

Let us now consider the MRA in more detail. One use for an MRA is to detect fluctuations that are transient in nature and limited to several scales. As one example, Figure 187 gives an expanded view of one portion of Figure 186 and indicates that a brief lull in variability occurred during December, 1985 (just to the left of the vertical dotted line marking the beginning of 1986). This lull occurs at short scales, namely, 1, 2 and 4 days (corresponding to $\tilde{D}_2$, $\tilde{D}_3$ and $\tilde{D}_4$). An examination of weather conditions during this period shows that the lull was due to the northward deflection of storms away from the Crescent City region by a high pressure ridge in the atmosphere that formed off the western United States during this time. Such ridges have slowly varying patterns of atmospheric pressure and wind patterns, as can be seen in the longer scales (8, 16 and 32 days, corresponding to $\tilde{D}_5$, $\tilde{D}_6$ and $\tilde{D}_7$). However, the ridge did not persist long enough to reverse the upward trend in seasonal water levels that is seen in the smooth $\tilde{S}_7$ (top series of Figure 187), even though both its high atmospheric pressure and northerly wind near the coast tend to produce lowered water levels. Its main influence on the water levels at Crescent City appears as a decrease in the variability at the shorter scales. In retrospect, this decrease can be seen in the plot of $\{X_t\}$ at the bottom of Figure 187, but it is very difficult to spot in that plot, whereas the MRA makes it easy to pick out this and other features of interest to oceanographers.

The MRA in Figure 186 also makes it clear that the variability at some scales is seasonally dependent. It is, however, problematic to quantitatively study the scale-dependent variance properties of $\{X_t\}$ from this analysis, because here

$$\|\mathbf{X}\|^2 \neq \sum_{j=1}^7 \|\tilde{D}_j\|^2 + \|\tilde{S}_7\|^2.$$

We can instead examine the MODWT coefficients, for which

$$\|\mathbf{X}\|^2 = \sum_{j=1}^7 \|\tilde{\mathbf{W}}_j\|^2 + \|\tilde{\mathbf{V}}_7\|^2.$$

We can qualitatively study the time dependence of the variance on a scale-by-scale basis by creating rotated cumulative variance plots, defined as follows. First we circularly advance the elements of $\widetilde{\mathbf{W}}_j$ by $|\nu_j^{(H)}|$ units so that $\mathcal{T}^{-|\nu_j^{(H)}|}\widetilde{\mathbf{W}}_j$ is aligned with $\mathbf{X}$ (see Equation (114b) and the discussion surrounding it). We then cumulatively sum the squares of the elements of $\mathcal{T}^{-|\nu_j^{(H)}|}\widetilde{\mathbf{W}}_j$ and normalize by $1/N$ to form

$$C_{j,t} \equiv \frac{1}{N} \sum_{u=0}^t \widetilde{\mathbf{W}}_{j,u+|\nu_j^{(H)}| \bmod N}^2, \quad t = 0, \dots, N-1.$$

Since $C_{j,N-1} = \frac{1}{N} \|\mathcal{T}^{-|\nu_j^{(H)}|}\mathbf{W}_j\|^2 = \frac{1}{N} \|\mathbf{W}_j\|^2$ can be regarded as the contribution to the sample variance of $\mathbf{X}$ due to physical scale $\tau_j \Delta t$, the series $\{C_{j,t}\}$ tracks the build up of the sample variance for the j th scale across time. Because $\{C_{j,t}\}$ is a nondecreasing sequence, it is sometimes difficult to deduce variations in the manner in which the sample variance is building up by plotting $C_{j,t}$ versus t , but we can overcome this difficulty by instead plotting

$$C'_{j,t} \equiv C_{j,t} - t \frac{C_{j,N-1}}{N-1} \quad \text{versus } t, \quad (189)$$

which we refer to as a rotated cumulative variance plot (the rotation essentially allows us to see deviations from a uniform build up of $\frac{C_{j,N-1}}{N-1}$ per increment in t ; this idea is exploited again in Section 9.6 to estimate the location of a change in variance in a time series).

As an example, Figure 190 shows rotated cumulative variance plots for physical scales $\tau_2 \Delta t = 1$ day and $\tau_7 \Delta t = 32$ days (since $\nu = -3$ for the LA(8) wavelet filter, Equation (114b) tells us that $\nu_2^{(H)} = -11$ and $\nu_7^{(H)} = -445$). The bottom plot in this figure clearly indicates that the variability in changes on a scale of a day is seasonally dependent, with higher variability occurring during the winter (i.e., close to the vertical dotted lines). The linear appearance of the rotated cumulative variance plot during the summer months shows that the variability at mid-summer is quite stable and predictable from year to year. The point at which the stable summer pattern shifts into the more erratic winter pattern can change from one year to the next by up to a month; moreover, the shift appears to be fairly abrupt some years (1985) and diffuse in others (1988). On the other hand, the top plot of Figure 190 shows a corresponding plot for variability in changes on a scale of 32 days. The seasonal dependence at this scale is much more erratic. For example, the rotated cumulative variance indicates a marked increase in variability around the beginning of 1980, 1982, 1983, 1986, 1987 and 1991, but quite constant variability of a roughly 20 month span from about March 1989 to January 1991. (We consider the time varying aspect of this rotated cumulative variance further in Section 8.7.)

Finally, as we noted in Section 5.3, the details and smooths constructed by the MODWT can be regarded as outputs from zero phase filters. This zero phase property is important if we want to relate events in the details to events in the original series and if we want to understand the temporal relationships between events at different scales. To illustrate what this property of zero phase means in practice, we show in the top plot of Figure 191 portions of the MODWT detail $\widetilde{\mathcal{D}}_5$ (thin curve) and $\mathbf{X}$ (thick) around the first part of 1986. This detail should reflect changes in $\mathbf{X}$ on a scale of

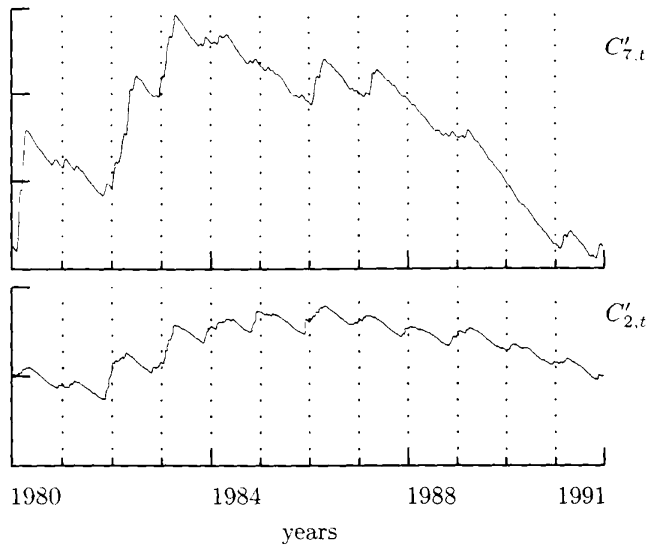


Figure 190. Rotated cumulative variance plots for subtidal sea level variations at physical scales $\tau_2 \Delta t = 1$ day (bottom plot) and $\tau_7 \Delta t = 32$ days (top).

8 days, so we would expect values in $\tilde{\mathcal{D}}_5$ to be large when an average over roughly 8 days differs markedly from values surrounding it. This time period was picked because there are three peaks in $\mathbf{X}$ of such a duration – these are centered at about 32, 48 and 72 days from the beginning of 1986 (note that the distance between minor tick marks on the horizontal axis is 8 days). The MODWT detail picks out these events nicely, and the locations of the peaks in $\tilde{\mathcal{D}}_5$ roughly match those in $\mathbf{X}$. Because $\tilde{\mathcal{D}}_5$ circularly shifts when we apply circular shifts to $\mathbf{X}$, the top plot will remain the same no matter what origin we pick for $\mathbf{X}$.

By way of contrast, the thin curves in the middle and bottom plots of Figure 191 show DWT details $\mathcal{D}_5$ for two different shifts in the origin of $\mathbf{X}$, namely, 12 and 8 values (corresponding to 6 and 4 days). Note that changing the origin of $\mathbf{X}$ by 2 days markedly changes the $\mathcal{D}_5$'s. Because DWT details do not possess the zero phase property, the locations of peaks in $\mathcal{D}_5$ need not match up very well with those in $\mathbf{X}$. For example, consider the location of the peak in $\mathcal{D}_5$ corresponding to the one in $\mathbf{X}$ near 32 days. Relative to $\tilde{\mathcal{D}}_5$ in the upper plot, the peak in $\mathcal{D}_5$ occurs markedly later in the middle plot and markedly earlier in the bottom plot. The ability of DWT details to locate transient events in $\mathbf{X}$ properly is thus limited, whereas MODWT details perform well because of their zero phase property.

5.9 Example: Nile River Minima

Here we present an MRA of a time series consisting of measurements of the minimum yearly water level of the Nile River over the years 622 to 1284 as recorded at the Roda gauge near Cairo (this analysis is adapted from an unpublished 1995 report by Simon Byers, Department of Statistics, University of Washington). The time series, with water levels in meters, is plotted in the bottom of Figure 192 and is actually a portion

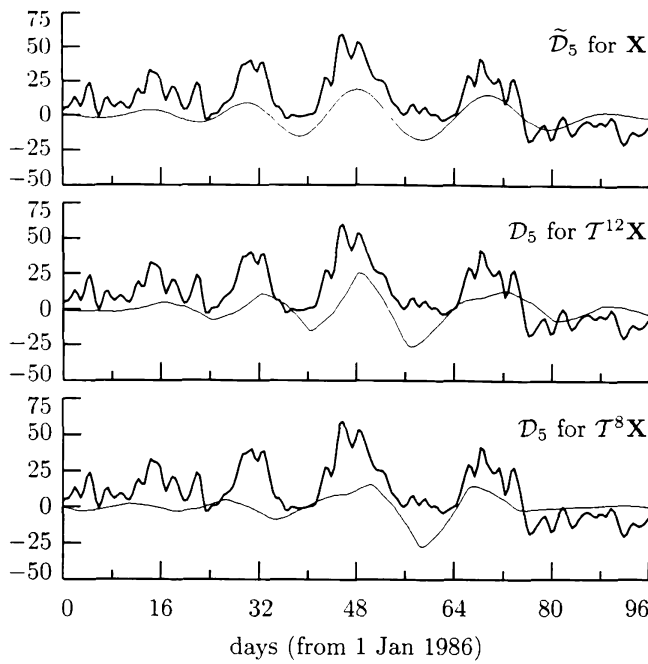


Figure 191. Demonstration of zero phase property of MODWT details. The thick curve in each plot is a 96 day portion of the Crescent City series starting at the beginning of 1986. The thin curves (from top to bottom) are the LA(8) MODWT detail $\tilde{D}_5$ for the original time series $\mathbf{X}$, the LA(8) DWT detail D_5 for $T^{12}\mathbf{X}$ (the time series delayed 12 data values) and the LA(8) DWT detail D_5 for $T^8\mathbf{X}$ (the time series delayed 8 data values). The physical scale associated with these details is $\tau_5 \Delta t = 8$ days; i.e., the details should be associated with changes of averages on a scale of 8 days (this is the distance between minor tick marks on the horizontal axis).

of a much longer historical record of Nile River minima compiled from various sources by Prince Omar Toussoun and published as part of a lengthy monogram entitled *Mémoire sur L'Histoire du Nil* in multiple volumes of *Mémoires de l'Institut d'Égypte* in 1925. Toussoun's record extends up to 1921, with the years 622 to 1284 representing the longest segment without interruptions (after 1284 there are a few scattered missing values up until 1470, after which coverage is much less complete: for example, there are only 17 measurements during the entire sixteenth century). The building of the Aswan Dam (1902) and the Aswan High Dam (1960–70) ended the primary role of nature in controlling the flow of the Nile River.

While the phenomena primarily responsible for the time series in Figure 192 are no longer operable, the historical record of Nile River minima has developed an interesting history in its own right. The hydrologist Hurst (1951) used this and other time series in a study that demonstrated the ineffectiveness of traditional statistical methodology for time series exhibiting long term persistence (i.e., a slowly decaying autocorrelation sequence – see Section 7.4). After Hurst's pioneering work, Mandelbrot and Wallis (1968) proposed to account for Hurst's empirical results by modeling the various series he studied as fractional Gaussian noise (this is defined in Section 7.6).

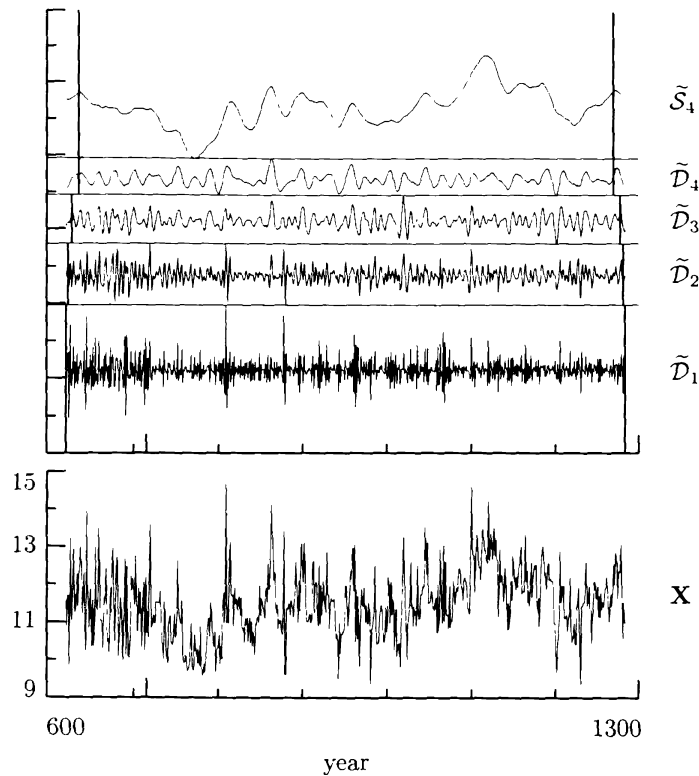


Figure 192. Level $J_0 = 4$ MODWT multiresolution analysis of Nile River minima (in meters) using the Haar wavelet filter.

This model was the first thorough attempt at characterizing what are now known as stationary long memory processes (Chapter 9 looks at various ways that wavelets can be used to study such processes). Beran (1994) used the Nile River series extensively as an example in his pioneering book on the statistics of long memory processes. More recently Eltahir and Wang (1999) studied this series as an indirect means of determining how often the El Niño phenomenon has occurred historically.

The top portion of Figure 192 shows a MODWT MRA of level $J_0 = 4$ for the Nile River minima. We used the Haar wavelet filter because MRAs from filters with $L = 4$ or larger were qualitatively similar to the Haar MRA – since there was no discernible benefit in using the wider filters, the Haar filter is preferable here because it minimizes concerns about the effects of boundary conditions (we have marked the boundary regions of each MRA component via a pair of thick vertical lines – see Section 5.11). We chose $J_0 = 4$ because we are primarily interested in pointing out a certain time dependency across scales τ_1 and τ_2 and contrasting it to a lack of such dependency at higher scales. Since the data were recorded annually (i.e., $\Delta t = 1$ year), the components of the MRA are associated with, from bottom to top, changes in averages on physical scales of 1, 2, 4 and 8 years ($\tilde{D}_1$ up to $\tilde{D}_4$, respectively) and averages over 16 years ($\tilde{S}_4$). The most interesting aspect of the MRA is that

inhomogeneity of variance in details $\tilde{D}_1$ and $\tilde{D}_2$: visually it appears that there is greater variability in these two details up until the early 700s. The details for higher scales ($\tilde{D}_3$ and $\tilde{D}_4$) show no evidence of increased variability during the early part of the record. While the increased variability is apparent – to a somewhat lesser degree – in the plot of $\{X_t\}$ itself and has been investigated using other methods (Beran, 1994), wavelet-based MRA picks it out quite clearly and nicely demonstrates that it is localized in scale. An examination of Toussoun's monogram and subsequent historical studies by Popper (1951) and Balek (1977) all indicate the construction in 715 (or soon thereafter) of a 'nilometer' in a mosque on Roda Island in the Nile River near Cairo (this year is marked by an elongated tick mark on the horizontal axes in both plots of Figure 192). The yearly minimum water levels from then to 1284 were measured using this device (or a reconstruction of it done in 861). The precise source of measurements for 622 to 714 is unknown, but they were most likely made at different locations around Cairo with possibly different types of measurement devices of less accuracy than the one in the Roda Island mosque. It is thus quite reasonable that this new nilometer led to a reduction in variability at the very smallest scales (it would be difficult to explain this change in variability on a geophysical basis).

We return to this example in Sections 8.8 and 9.8, where we make use of informal and formal statistical tests to verify that the variability is significantly different at scales of 1 and 2 days before and after about 715.

5.10 Example: Ocean Shear Measurements

Here we examine a 'time' series of vertical ocean shear measurements. These data were collected by an instrument that is dropped over the side of a ship and designed then to descend vertically into the ocean. As it descends, the probe collects measurements concerning the ocean as a function of depth. The ordering variable of our 'time' series is thus depth. One of the measurements is the x component of the velocity of water. This velocity is collected every 0.1 meters, first differenced over an interval of 10 meters, and then low-pass filtered to obtain a series related to vertical shear in the ocean. The bottom panel of Figure 194 shows the series of vertical shear measurements used here (this series has units of inverse seconds). The data extend from a depth of 350.0 meters down to 1037.4 meters in increments of $\Delta t = 0.1$ meters (there are $N = 6875$ data values in all). There are two thin vertical lines marked on the plot, between which there are 4096 values ranging from 489.5 meters to 899.0 meters (this subseries is discussed further in Section 8.9).

Figure 194 shows an MRA of level $J_0 = 6$ constructed using the LA(8) wavelet filter. We chose this filter primarily because, when we return to this example in Section 8.9, we will be interested both in looking at how the wavelet coefficients at different scales are related across time and also in using a wavelet filter that offers good protection against leakage (the concept of leakage is discussed in Section 8.9 also). The natural choices to achieve alignment in time are the least asymmetric or coiflet filters. Within these two classes, the LA(8) is the shortest filter with acceptable leakage properties (had we been interested solely in an MRA, the D(4), D(6) and C(6) filters would yield MRAs that are quite comparable to that of the LA(8) filter, but the small scale details $\tilde{D}_j$ for the Haar MRA look considerably different due to leakage). We set $J_0 = 6$ because this is the index of the smallest scale that is visibly free of bursts (to be discussed in what follows). This choice yields an MRA that isolates the

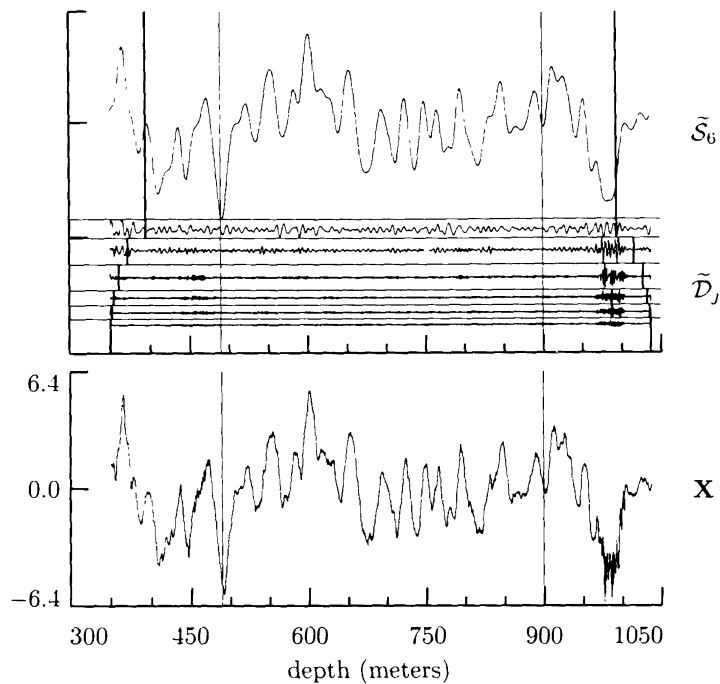


Figure 194. Level $J_0 = 6$ MODWT multiresolution analysis using LA(8) wavelet of vertical shear measurements (in inverse seconds) versus depth (in meters). This series was collected and supplied by Mike Gregg, Applied Physics Laboratory, University of Washington.

series that is burst free.

The MRA in Figure 194 is dominated by the smooth $\tilde{S}_6$. This smooth appears to be a reasonable extraction of the large scale variations in $\{X_t\}$ except at the very beginning and end of the series, where the periodic assumption inherent in the MODWT arguably causes distortions in $\tilde{S}_6$ (as before, the pair of thick vertical lines intersecting a given $\tilde{D}_j$ or $\tilde{S}_6$ delineates the portions of the MRA influenced at least in part by boundary conditions). If so desired, we could use reflection boundary conditions to alleviate these slight distortions (see the discussion surrounding Equation (140)). Because the smooth dominates the MRA to the point that the details are not distinguishable, we have plotted the $\tilde{D}_j$'s separately in Figure 195. The details nicely pick out two bursts, one near 450 meters, and the other just after 950 meters (the second burst is clearly visible in the plot of $\{X_t\}$ in Figure 194, but the first is only apparent in retrospect). Since the physical scale associated with $\tilde{D}_j$ is $\tau_j \Delta t = 0.1\tau_j$ meters, we see that the first burst is prominent at scales of 0.8 meters and smaller (i.e., $\tilde{D}_1$ up to $\tilde{D}_4$), while there is still prominent evidence of the second burst at scale 1.6 meters (i.e., $\tilde{D}_5$). An examination of MRAs for $J_0 \geq 7$ shows no indication of these bursts for physical scales longer than 3.2 meters. If we examine the first burst more closely, we see that it shuts down rapidly in $\tilde{D}_4$ (0.8 meters), but the burst continues beyond this shut down point in smaller scale details ($\tilde{D}_3$, $\tilde{D}_2$ and $\tilde{D}_1$). A physical interpretation might be that turbulence at a 0.8 meter scale is driving shorter scale turbulence at

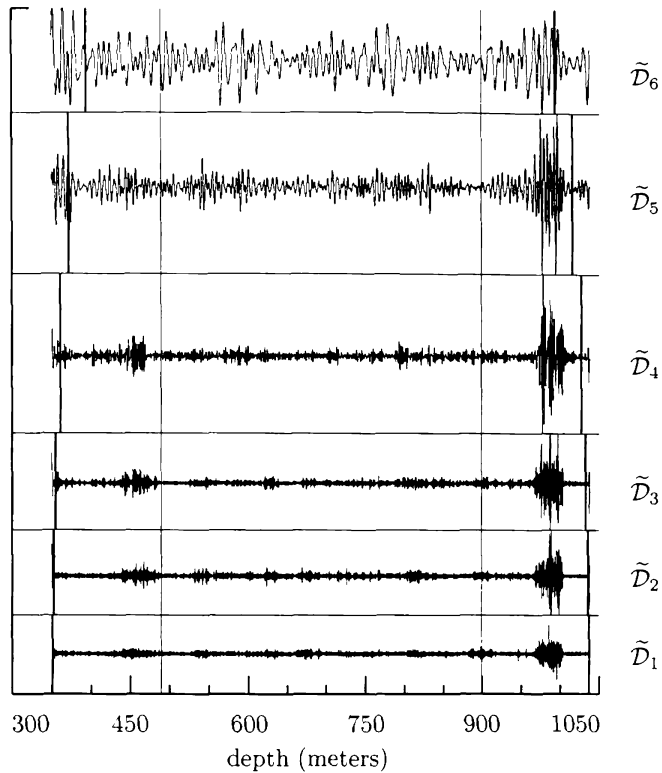


Figure 195. Expanded view of details series in MRA shown in Figure 194.

greater depths. By contrast, the larger second burst seems to be colocated in details $\tilde{D}_1$ up to $\tilde{D}_4$, but there seems to be hints of increased variability in $\tilde{D}_5$ and $\tilde{D}_6$ prior to the main portion of the burst. A MODWT MRA facilitates studies such as this one in which the depth (or time) relationships between events at different scales are important because its details and smooth are associated with zero phase filters. (While the additive decomposition given by the MRA gives a useful basis description of the shear series, we can say more about its properties in Section 8.9 after we have discussed ways of assessing a wavelet-based ANOVA with simple statistical models.)

5.11 Practical Considerations

In Section 4.11 we discussed several practical considerations that must be taken into account when carrying out a DWT analysis of a time series. Some of the remarks made there are also appropriate for the MODWT; however, because there are significant differences between the DWT and MODWT, each of the following topics merits more discussion.

- *Choice of wavelet filter*

For the most part, the remarks made in Section 4.11 about selecting a particular wavelet filter for the DWT hold also for the MODWT, but use of the MODWT for

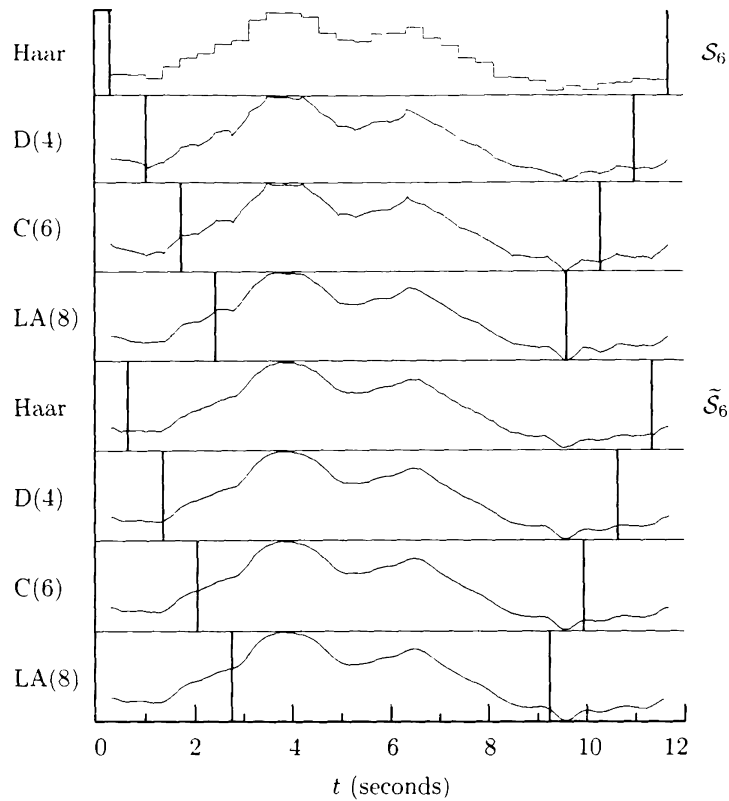


Figure 197. Comparison of DWT smooths $\mathcal{S}_6$ (top four series) and MODWT smooths $\tilde{\mathcal{S}}_6$ (bottom four) for the ECG time series using, from top to bottom within each group, the Haar, D(4), C(6) and LA(8) wavelet filters. The thick vertical lines mark the boundary regions of the smooths, i.e., the parts of the smooths that are influenced (at least to some degree) by the assumption of circularity.

creating an MRA makes the choice of the wavelet filter less critical, as we illustrate in Figure 197. The top four series there show the DWT-based smooths $\mathcal{S}_6$ from the Haar, D(4), C(6) and LA(8) MRAs for the ECG time series (these are taken from Figures 130 to 133). The smooths based on the Haar, D(4) and C(6) wavelet filters contain unappealing blocks, sharks' fins and triangles, which are artifacts due to the filters rather than pertinent features in the ECG data. On the other hand, the LA(8) smooth shows no obvious artifacts and hence can be regarded as a more realistic depiction of the baseline drift in this series. These considerations would lead us to recommend against using Haar, D(4) or C(6) DWT-based MRAs for most time series since there are not many series whose characteristic features are well matched to blocks, sharks' fins and triangles.

Let us now see what happens when we switch from the DWT to the MODWT. The bottom four series in Figure 197 show MODWT smooths $\tilde{\mathcal{S}}_6$ based upon the same four wavelet filters as before. Qualitatively, the three smooths look quite similar, although the Haar $\tilde{\mathcal{S}}_6$ shows evidence of very small ripples that are not present in the

other three MODWT smooths. A similar examination of the MODWT details $\tilde{D}_j$ for the ECG series again shows only quite small differences due to the different wavelet filters. We thus cannot as easily dismiss the Haar, D(4) and C(6) MRAs as we could in the DWT case. In general, we must resort to application-dependent considerations in order to make a reasonable choice of wavelet filters (for example, if $\tilde{S}_6$ is meant to capture the baseline drift and if we assume for the sake of argument that the notion of this drift precludes small ripples, an ECG analyst might prefer the D(4), C(6) or LA(8) wavelet filters over the Haar).

The key to understanding why MODWT-based MRAs are less dependent on the wavelet filter than DWT-based MRAs is given in Exercise [5.1], which is to show that MODWT details and smooths can be generated by averaging certain circularly shifted DWT details and smooths created from circularly shifted versions of a time series. This extra layer of averaging acts to blur out blocks, sharks' fins and triangles, thus eliminating the obvious defects in the DWT-based MRAs. From the point of view of filtering theory, the desirable properties of the MODWT detail $\tilde{D}_j$ arise because it is the output from a filter cascade with two stages (see Section 5.4). The first stage involves an 'analysis' filter with transfer function $\tilde{H}_j(\cdot)$, and the second stage, a 'synthesis' filter $\tilde{H}_j^*(\cdot)$. The equivalent filter $|\tilde{H}_j(\cdot)|^2$ has a phase function that is always zero, no matter what wavelet filter we choose to use. This interpretation suggests that we can expect to find more marked differences between MODWT wavelet coefficients $\tilde{W}_j$ from different wavelet filters than between the corresponding details: the $\tilde{W}_j$ are formed using $\tilde{H}_j(\cdot)$, whose phase properties depend upon the particular wavelet filter. Additionally, the squared gain functions associated with the MODWT wavelet coefficients and details are, respectively, $|\tilde{H}_j(\cdot)|^2$ and $|\tilde{H}_j(\cdot)|^4$, so we can expect the latter to provide a decent approximation to a band-pass filter in certain cases where the former might be inadequate. This point is illustrated by the example in Section 8.9, where we show that use of the Haar $\tilde{W}_j$ leads to an ANOVA that suffers from 'leakage' when compared to $\tilde{W}_j$ based on wavelet filters of length $L \geq 4$.

To summarize, as compared to DWT-based MRAs, a MODWT-based MRA is less dependent upon our choice of wavelet filter, but not so much so that we can recommend always using a particular filter. A careful study of the differences between MRAs based on different wavelet filters is still needed to see which filter is best matched to a particular application. We can expect the wavelet filter to have a greater impact on the MODWT wavelet coefficients because these depend more highly on the characteristics of individual wavelet filters.

- *Handling boundary conditions*

Like the DWT, the MODWT makes use of circular filtering, so the comments we made in Section 4.11 about circular boundary conditions and the simple alternative of reflection boundary conditions apply equally well to the MODWT. In what follows, we consider the extent to which circularity affects the MODWT coefficients and MRAs – our discussion closely parallels the one for the DWT, but the relevant formulae are quite different.

Let us first determine exactly which MODWT coefficients are affected by the assumption of circularity. To do so, we just have to look at the definitions for $\tilde{W}_{j,t}$ and $\tilde{V}_{j,t}$ in Equation (169a) and see for which values of t we can eliminate the 'mod' operation in $X_{t-l \bmod N}$ without changing the way in which these coefficients are defined. In forming either coefficient, the dummy index l ranges from 0 to $L_j - 1 = (2^j - 1)(L - 1)$.

Since $t - l \bmod N = t - l$ for all l as long as $t \geq L_j - 1$, the MODWT coefficients that are affected by circularity are

$$\tilde{W}_{j,t} \text{ and } \tilde{V}_{j,t} \text{ for } t = 0, \dots, \min\{L_j - 2, N - 1\}. \quad (198a)$$

We refer to the above as the MODWT boundary coefficients. As was true for the DWT coefficients in $\mathbf{W}_j$ and $\mathbf{V}_j$, the boundary coefficients all occur at the beginning of $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_j$. Since the dimensions of these vectors are always N and since L_j increases with j , the proportion of boundary coefficients increases with j and reaches unity when $L_j - 1 \geq N$, i.e., when $j \geq \log_2(\frac{N}{L-1} + 1)$.

Let us now specialize to the LA and coiflet wavelet filters and assume that $L_j \leq N$ so that $\min\{L_j - 2, N - 1\} = L_j - 2$ (i.e., we have at least one coefficient that is not a boundary coefficient). The discussion in Sections 4.8 and 4.9 says that, in order to obtain filter outputs that are aligned with $\mathbf{X}$, we must circularly shift the MODWT coefficient vectors by amounts that are dictated by the phase properties of the LA or coiflet filters. With $\nu_j^{(H)}$ and $\nu_j^{(G)}$ defined as in Equations (114b) and (114a), this procedure yields the circularly shifted vectors $\mathcal{T}^{-|\nu_j^{(H)}|}\tilde{\mathbf{W}}_j$ and $\mathcal{T}^{-|\nu_j^{(G)}|}\tilde{\mathbf{V}}_j$, whose elements can then be plotted versus the same actual times associated with the corresponding elements of $\mathbf{X}$. Thus some of the MODWT boundary coefficients will be transferred to the ends of the circularly shifted vectors. The indices of the boundary wavelet coefficients in $\mathcal{T}^{-|\nu_j^{(H)}|}\tilde{\mathbf{W}}_j$ are given by

$$t = 0, \dots, L_j - 2 - |\nu_j^{(H)}| \text{ along with } t = N - |\nu_j^{(H)}|, \dots, N - 1: \quad (198b)$$

likewise, the indices of the boundary coefficients in $\mathcal{T}^{-|\nu_j^{(G)}|}\tilde{\mathbf{V}}_j$ are given by

$$t = 0, \dots, L_j - 2 - |\nu_j^{(G)}| \text{ along with } t = N - |\nu_j^{(G)}|, \dots, N - 1 \quad (198c)$$

(Exercise [5.11] is to check that one aspect of these equations is sensible, namely, that we always have $L_j - 2 \geq |\nu_j^{(H)}|$ and $L_j - 2 \geq |\nu_j^{(G)}|$). These results can be used to add vertical lines to plots of the circularly shifted vectors, with the interpretation that these lines delineate the portions of the plotted coefficients outside of which circularity has some effect. As an example, these lines appear as part of Figure 183 showing the LA(8) MODWT of the ECG series.

Finally let us study how circularity affects the MODWT details $\tilde{\mathcal{D}}_j$ and smooths $\tilde{\mathcal{S}}_j$ (this discussion applies to all wavelet filters: i.e., we are no longer restricting ourselves to the LA or coiflet filters). As indicated by Equation (172a), the detail $\tilde{\mathcal{D}}_j$ can be obtained directly from $\mathbf{X}$ by filtering the latter with a circular filter given by $\{|\tilde{H}_j(\frac{k}{N})|^2\}$. This filter is a periodized filter formed by the cross-correlation of the filter $\{\tilde{h}_{j,l}\}$ with itself, i.e., the autocorrelation of $\{\tilde{h}_{j,l}\}$ (see Equation (36e)). If we denote this filter by $\{\tilde{h}_j \star \tilde{h}_{j,l}\}$, we can use Equation (36c) to write

$$\tilde{h}_j \star \tilde{h}_{j,l} = \sum_{u=-\infty}^{\infty} \tilde{h}_{j,u} \tilde{h}_{j,u+l} = \sum_{u=0}^{L_j-1} \tilde{h}_{j,u} \tilde{h}_{j,u+l}, \quad l = \dots, -1, 0, 1, \dots$$

where we have used the fact that $\tilde{h}_{j,l} = 0$ for $l < 0$ and $l \geq L_j$ (in comparing the above to (36c), recall that $\tilde{h}_{j,l}$ is real-valued and hence $\tilde{h}_{j,l}^* = \tilde{h}_{j,l}$). From the above,

we can argue that $\tilde{h}_j \star \tilde{h}_{j,l} = 0$ when $|l| \geq L_j$; i.e., the filter $\{\tilde{h}_j \star \tilde{h}_{j,l}\}$ is of odd width $2L_j - 1$ and is centered about $l = 0$. We can thus write the elements of $\tilde{\mathcal{D}}_j$ as

$$\tilde{\mathcal{D}}_{j,t} = \sum_{l=-(L_j-1)}^{L_j-1} \tilde{h}_j \star \tilde{h}_{j,l} X_{t-l \bmod N}, \quad t = 0, \dots, N-1,$$

from which it is evident that, as long as we have both $t - (L_j - 1) \geq 0$ and $t + (L_j - 1) \leq N - 1$, the element $\tilde{\mathcal{D}}_{j,t}$ is unaffected by circularity; i.e., we can write

$$\tilde{\mathcal{D}}_{j,t} = \sum_{l=-(L_j-1)}^{L_j-1} \tilde{h}_j \star \tilde{h}_{j,l} X_{t-l} \text{ for } t = L_j - 1, \dots, N - L_j.$$

The elements of $\tilde{\mathcal{D}}_j$ that are affected by circularity are thus those with indices t satisfying either $t = 0, \dots, L_j - 2$ or $t = N - L_j + 1, \dots, N - 1$. The same argument also holds for $\tilde{\mathcal{S}}_j$.

We can use the above results to place vertical lines on plots of $\tilde{\mathcal{D}}_j$ and $\tilde{\mathcal{S}}_j$ to delineate the boundary regions, i.e., the regions influenced by circularity. This has been done in Figures 183, 184, 186, 192, 194 and 195 and also on the MODWT smooths $\tilde{\mathcal{S}}_6$ plotted in Figure 197 (these are the bottom four series). From this latter figure, we see that, as the wavelet filter width L increases, the extent of the boundary region increases. For Figure 197 the boundary regions are also delineated on the corresponding DWT smooths $\mathcal{S}_6$ (the top four plotted series). A comparison between $\mathcal{S}_6$ and $\tilde{\mathcal{S}}_6$ for a given wavelet filter shows that the MODWT boundary regions are somewhat larger than the DWT regions, a price that we must unavoidably pay for the appealing properties of MODWT MRAs. As we emphasized in Section 4.11, we must regard the boundary regions as very conservative measures of the influence of circularity – the regions over which circularity has a deleterious effect are usually much smaller.

- *Handling sample sizes that are not a power of two*

As we have noted earlier in this chapter, the MODWT is defined naturally for all sample sizes. Unlike the DWT, there is thus no need for special adaptations to handle certain sample sizes.

- *Choice of level J_0 of MODWT*

Our discussion on selecting J_0 for the DWT noted that a sensible choice depends primarily on the time series at hand – the same also holds for the MODWT. It is interesting to note, however, that, if $N = 2^J$, we cannot set J_0 to be greater than J for the DWT (because the DWT pyramid algorithm must terminate when there is but a single scaling coefficient left), whereas in theory there is nothing to stop us from picking $J_0 > J$ for the MODWT (because the MODWT pyramid algorithm always yields N scaling coefficients for use at the next level). Setting $J_0 > J$ would superficially seem to offer a way of obtaining information about variations in our times series over scales greater than the entire extent of the series, which is counter-intuitive. Exercise [5.9c] indicates that in fact we gain nothing by setting $J_0 > J$: when $N = 2^J$, all the elements of $\tilde{\mathcal{S}}_{J_0}$ must be equal to the sample mean $\bar{X}$ for all $J_0 \geq J$. Since Equation (172c) says that $\tilde{\mathcal{D}}_{J_0+1} = \tilde{\mathcal{S}}_{J_0} - \tilde{\mathcal{S}}_{J_0+1}$, all MODWT details for levels $J_0 > J$

are constrained to be zero. For sample sizes that are not a power of two, in practice $\tilde{D}_{J_0}$ rapidly becomes quite close to zero as $2^{J_0}/N$ increases beyond about 1.5.

If, as in the DWT discussion, we insist that there be at least one MODWT coefficient in $\tilde{\mathbf{V}}_{J_0}$ that is not a boundary coefficient, then a rule of thumb is to require the condition $J_0 < \log_2(\frac{N}{L-1} + 1)$; if this rule is deemed to be too conservative, alternative upper bounds are $J_0 \leq \log_2(N)$ or, in rare cases, $J_0 \leq \log_2(1.5N)$. All three upper bounds should be regarded as just guides for initially picking J_0 – the final choice of J_0 must come from application-dependent considerations.

5.12 Summary

Let $\mathbf{X}$ be an N dimensional vector whose elements are the real-valued time series $\{X_t : t = 0, \dots, N-1\}$, where the sample size N is any positive integer. For any positive integer J_0 , the level J_0 MODWT of $\mathbf{X}$ is a transform consisting of the $J_0 + 1$ vectors $\tilde{\mathbf{W}}_1, \dots, \tilde{\mathbf{W}}_{J_0}$ and $\tilde{\mathbf{V}}_{J_0}$, all of which have dimension N . The vector $\tilde{\mathbf{W}}_j$ contains the MODWT wavelet coefficients associated with changes on scale $\tau_j \equiv 2^{j-1}$, while $\tilde{\mathbf{V}}_{J_0}$ contains the MODWT scaling coefficients associated with averages on scale $\lambda_{J_0} \equiv 2^{J_0}$. Conceptually, based upon the definition for the MODWT coefficients (see below), we can write $\tilde{\mathbf{W}}_j = \tilde{\mathcal{W}}_j \mathbf{X}$ and $\tilde{\mathbf{V}}_{J_0} = \tilde{\mathcal{V}}_{J_0} \mathbf{X}$, where $\tilde{\mathcal{W}}_j$ and $\tilde{\mathcal{V}}_{J_0}$ are $N \times N$ matrices (in practice, however, the MODWT coefficient vectors are generated via an efficient ‘pyramid’ algorithm summarized below). The time series $\mathbf{X}$ can be recovered from its MODWT via

$$\mathbf{X} = \sum_{j=1}^{J_0} \tilde{\mathcal{W}}_j^T \tilde{\mathbf{W}}_j + \tilde{\mathcal{V}}_{J_0}^T \tilde{\mathbf{V}}_{J_0} \equiv \sum_{j=1}^{J_0} \tilde{\mathcal{D}}_j + \tilde{\mathcal{S}}_{J_0},$$

which defines a MODWT-based multiresolution analysis (MRA) of $\mathbf{X}$ in terms of j th level MODWT details $\tilde{\mathcal{D}}_j \equiv \tilde{\mathcal{W}}_j^T \tilde{\mathbf{W}}_j$ and the J_0 level MODWT smooth $\tilde{\mathcal{S}}_{J_0} \equiv \tilde{\mathcal{V}}_{J_0}^T \tilde{\mathbf{V}}_{J_0}$. A scale-by-scale ANOVA can be based upon the energy decomposition

$$\|\mathbf{X}\|^2 = \sum_{j=1}^{J_0} \|\tilde{\mathbf{W}}_j\|^2 + \|\tilde{\mathbf{V}}_{J_0}\|^2.$$

Unlike the DWT, the MODWT is *not* an orthonormal transform, but it is nevertheless capable of producing the above MRA and energy decomposition, which are similar to what the DWT yields. A primary advantage of the MODWT is that, unlike the DWT, it is not as critically dependent upon the assumed starting point for a time series. In particular, if $\tilde{\mathbf{W}}_1, \dots, \tilde{\mathbf{W}}_{J_0}$ and $\tilde{\mathbf{V}}_{J_0}$ are the MODWT for $\mathbf{X}$, then the MODWT for the circularly shifted series $\mathcal{T}^m \mathbf{X}$ is given by $\mathcal{T}^m \tilde{\mathbf{W}}_1, \dots, \mathcal{T}^m \tilde{\mathbf{W}}_{J_0}$ and $\mathcal{T}^m \tilde{\mathbf{V}}_{J_0}$. This fact implies that the MRA for $\mathcal{T}^m \mathbf{X}$ has details $\mathcal{T}^m \tilde{\mathcal{D}}_j$ and smooth $\mathcal{T}^m \tilde{\mathcal{S}}_{J_0}$; on the other hand, the energy decomposition remains the same because $\|\mathcal{T}^m \tilde{\mathbf{W}}_j\|^2 = \|\tilde{\mathbf{W}}_j\|^2$ and $\|\mathcal{T}^m \tilde{\mathbf{V}}_{J_0}\|^2 = \|\tilde{\mathbf{V}}_{J_0}\|^2$.

By definition, the elements of $\tilde{\mathbf{W}}_j$ and $\tilde{\mathbf{V}}_{J_0}$ are outputs obtained by filtering $\mathbf{X}$, namely,

$$\tilde{W}_{j,t} \equiv \sum_{l=0}^{L_j-1} \tilde{h}_{j,l} X_{t-l \bmod N} \quad \text{and} \quad \tilde{V}_{J_0,t} \equiv \sum_{l=0}^{L_{J_0}-1} \tilde{g}_{J_0,l} X_{t-l \bmod N},$$

$t = 0, \dots, N-1$, where $\{\tilde{h}_{j,l}\}$ and $\{\tilde{g}_{J_0,l}\}$ are the j th level MODWT wavelet and scaling filters. These filters are defined in terms of the j th level equivalent wavelet and scaling

filters $\{h_{j,l}\}$ and $\{g_{j,l}\}$ for the DWT as $\tilde{h}_{j,l} = h_{j,l}/2^{j/2}$ and $\tilde{g}_{j,l} = g_{j,l}/2^{j/2}$. Each of the MODWT filters has width $L_j \equiv (2^j - 1)(L - 1) + 1$ and can be generated once a basic MODWT wavelet filter $\tilde{h}_{1,l} \equiv \tilde{h}_l \equiv h_l/\sqrt{2}$ and its related MODWT scaling filter $\tilde{g}_{1,l} \equiv \tilde{g}_l \equiv (-1)^{l+1}\tilde{h}_{L-1-l}$ have been specified. Specifically, if $\tilde{H}(\cdot)$ and $\tilde{G}(\cdot)$ are the transfer function for, respectively, $\{\tilde{h}_l\}$ and $\{\tilde{g}_l\}$, then the transfer functions for $\{\tilde{h}_{j,l}\}$ and $\{\tilde{g}_{j,l}\}$ are given by

$$\tilde{H}_j(f) \equiv \tilde{H}(2^{j-1}f) \prod_{l=0}^{j-2} \tilde{G}(2^l f) \quad \text{and} \quad \tilde{G}_j(f) \equiv \prod_{l=0}^{j-1} \tilde{G}(2^l f).$$

Because the MODWT filters are just rescaled versions of the corresponding equivalent filters for the DWT, all the relationships we have derived for the former hold for the latter after obvious adjustments – the most important ones are given in Table 202. If we let $\{\tilde{h}_{j,l}^\circ\}$ and $\{\tilde{g}_{j,l}^\circ\}$ be the periodization of $\{\tilde{h}_{j,l}\}$ and $\{\tilde{g}_{j,l}\}$ to circular filters of length N , we can use the definitions for the MODWT coefficients to obtain

$$\tilde{W}_{j,t} = \sum_{l=0}^{N-1} \tilde{h}_{j,l}^\circ X_{t-l \bmod N} \quad \text{and} \quad \tilde{V}_{j,t} = \sum_{l=0}^{N-1} \tilde{g}_{j,l}^\circ X_{t-l \bmod N}.$$

These can be used to deduce the elements of the matrices $\tilde{W}_j$ and $\tilde{V}_{j_0}$. In particular, Equation (171c) shows how each row of $\tilde{W}_j$ is constructed using the N elements of $\{\tilde{h}_{j,l}^\circ\}$ (the matrix $\tilde{V}_{j_0}$ is formed similarly using $\{\tilde{g}_{j,l}^\circ\}$ instead).

In practice the matrices $\tilde{W}_j$ and $\tilde{V}_{j_0}$ are not used explicitly, but rather the vectors constituting the MODWT are formed using a ‘pyramid’ algorithm that uses just the basic MODWT wavelet and scaling filters. The MODWT pyramid algorithm at stage j consists of circularly filtering the N elements of $\tilde{V}_{j-1} \equiv [\tilde{V}_{j-1,0}, \tilde{V}_{j-1,1}, \dots, \tilde{V}_{j-1,N-1}]^T$ to obtain the j th level MODWT wavelet and scaling coefficients via

$$\tilde{W}_{j,t} = \sum_{l=0}^{L-1} \tilde{h}_l \tilde{V}_{j-1,t-2^{j-1}l \bmod N} \quad \text{and} \quad \tilde{V}_{j,t} = \sum_{l=0}^{L-1} \tilde{g}_l \tilde{V}_{j-1,t-2^{j-1}l \bmod N}.$$

$t = 0, \dots, N-1$ (these are the elements of $\tilde{W}_j$ and $\tilde{V}_j$). With $\tilde{V}_0 \equiv \mathbf{X}$, we start the pyramid algorithm with $j = 1$ and, after repeating the algorithm with $j = 2, 3, \dots, J_0$, we obtain all $J_0 + 1$ of the MODWT coefficient vectors (the other vectors $\tilde{V}_1, \dots, \tilde{V}_{J_0-1}$ are MODWT scaling coefficients for scales smaller than λ_{J_0} and can be regarded as intermediate computations). The pyramid algorithm can be expressed in matrix notation as $\tilde{W}_j = \tilde{B}_j \tilde{V}_{j-1}$ and $\tilde{V}_j = \tilde{A}_j \tilde{V}_{j-1}$, where $\tilde{B}_j$ and $\tilde{A}_j$ are $N \times N$ matrices. The elements of these matrices are taken from circular filters of length N obtained by periodizing upsampled versions of $\{\tilde{h}_l\}$ and $\{\tilde{g}_l\}$. In particular, suppose we upsample $\{\tilde{h}_l\}$ by a factor of 2^{j-1} (i.e., we insert $2^{j-1} - 1$ zeros between each element of the filter). If we denote the result of periodizing this upsampled filter to length N as $\{\tilde{h}_l^{(\uparrow 2^{j-1})^\circ}\}$, then the first row of $\tilde{B}_j$ is given by

$$[\tilde{h}_0^{(\uparrow 2^{j-1})^\circ}, \tilde{h}_{N-1}^{(\uparrow 2^{j-1})^\circ}, \tilde{h}_{N-2}^{(\uparrow 2^{j-1})^\circ}, \dots, \tilde{h}_2^{(\uparrow 2^{j-1})^\circ}, \tilde{h}_1^{(\uparrow 2^{j-1})^\circ}],$$

and the $N-1$ remaining rows are obtained by circularly shifting the above to the right by, respectively, $1, 2, \dots, N-1$ units.

$\tilde{h}_l \equiv h_l/\sqrt{2}$	$\tilde{g}_l \equiv g_l/\sqrt{2}$
$\{\tilde{h}_l\} \longleftrightarrow \tilde{H}(\cdot) = \frac{1}{\sqrt{2}}H(\cdot)$	$\{\tilde{g}_l\} \longleftrightarrow \tilde{G}(\cdot) = \frac{1}{\sqrt{2}}G(\cdot)$
$\tilde{h}_l = (-1)^l \tilde{g}_{L-1-l}$	$\tilde{g}_l \equiv (-1)^{l+1} \tilde{h}_{L-1-l}$
$\tilde{H}(f) = -e^{-i2\pi f(L-1)} \tilde{G}(\frac{1}{2} - f)$	$\tilde{G}(f) = e^{-i2\pi f(L-1)} \tilde{H}(\frac{1}{2} - f)$
$\sum_l \tilde{h}_l = \tilde{H}(0) \equiv 0$	$\sum_l \tilde{g}_l = \tilde{G}(0) = 1$
$\sum_l \tilde{h}_l^2 \equiv \frac{1}{2}$	$\sum_l \tilde{g}_l^2 = \frac{1}{2}$
$\sum_l \tilde{h}_l \tilde{h}_{l+2n} \equiv 0, n \neq 0$	$\sum_l \tilde{g}_l \tilde{g}_{l+2n} = 0, n \neq 0$
$\sum_l \tilde{g}_l \tilde{h}_{l+2n} = 0$	
$\tilde{\mathcal{H}}(f) \equiv \tilde{H}(f) ^2 = \frac{1}{2}\mathcal{H}(f)$	$\tilde{\mathcal{G}}(f) \equiv \tilde{G}(f) ^2 = \frac{1}{2}\mathcal{G}(f)$
$\tilde{\mathcal{H}}(f) + \tilde{\mathcal{H}}(f + \frac{1}{2}) = 1$	$\tilde{\mathcal{G}}(f) + \tilde{\mathcal{G}}(f + \frac{1}{2}) = 1$
$\tilde{\mathcal{G}}(f) + \tilde{\mathcal{H}}(f) = 1$	
$\tilde{W}_{1,t} \equiv \sum_l \tilde{h}_l X_{t-l \bmod N}$	$\tilde{V}_{1,t} \equiv \sum_l \tilde{g}_l X_{t-l \bmod N}$
$\tilde{W}_{j,t} = \sum_l \tilde{h}_l \tilde{V}_{j-1,t-2^{j-1}l \bmod N}$	$\tilde{V}_{j,t} = \sum_l \tilde{g}_l \tilde{V}_{j-1,t-2^{j-1}l \bmod N}$
$\tilde{h}_{1,l} \equiv \tilde{h}_l, \quad \tilde{H}_1(f) \equiv \tilde{H}(f)$	$\tilde{g}_{1,l} \equiv \tilde{g}_l, \quad \tilde{G}_1(f) \equiv \tilde{G}(f)$
$\tilde{H}_j(f) \equiv \tilde{H}(2^{j-1}f) \prod_{l=0}^{j-2} \tilde{G}(2^l f)$	$\tilde{G}_j(f) \equiv \prod_{l=0}^{j-1} \tilde{G}(2^l f)$
$\tilde{H}_j(f) = \frac{1}{2^{j/2}} H_j(f)$	$\tilde{G}_j(f) = \frac{1}{2^{j/2}} G_j(f)$
$\tilde{H}_j(f) = \tilde{H}(2^{j-1}f) \tilde{G}_{j-1}(f)$	$\tilde{G}_j(f) = \tilde{G}(2^{j-1}f) \tilde{G}_{j-1}(f)$
$\{\tilde{h}_{j,l}\} \longleftrightarrow \tilde{H}_j(\cdot)$	$\{\tilde{g}_{j,l}\} \longleftrightarrow \tilde{G}_j(\cdot)$
$\sum_l \tilde{h}_{j,l} = \tilde{H}_j(0) = 0$	$\sum_l \tilde{g}_{j,l} = \tilde{G}_j(0) = 1$
$\sum_l \tilde{h}_{j,l}^2 = \frac{1}{2^j}$	$\sum_l \tilde{g}_{j,l}^2 = \frac{1}{2^j}$
$\sum_l \tilde{h}_{j,l} \tilde{h}_{j,l+2^j n} = 0, n \neq 0$	$\sum_l \tilde{g}_{j,l} \tilde{g}_{j,l+2^j n} = 0, n \neq 0$
$\sum_l \tilde{g}_{j,l} \tilde{h}_{j,l+2^j n} = 0$	
$\tilde{\mathcal{H}}_j(f) \equiv \tilde{H}_j(f) ^2 = \frac{1}{2^j} \mathcal{H}_j(f)$	$\tilde{\mathcal{G}}_j(f) \equiv \tilde{G}_j(f) ^2 = \frac{1}{2^j} \mathcal{G}_j(f)$
$\tilde{W}_{j,t} \equiv \sum_l \tilde{h}_{j,l} X_{t-l \bmod N}$	$\tilde{V}_{j,t} \equiv \sum_l \tilde{g}_{j,l} X_{t-l \bmod N}$

Table 202. Key relationships involving MODWT wavelet and scaling filters (the conventions in Table 154 for limits on sums over l apply here too).

Given $\widetilde{\mathbf{W}}_j$ and $\widetilde{\mathbf{V}}_j$, we can recover the elements of $\widetilde{\mathbf{V}}_{j-1}$ via the j th stage of the inverse MODWT pyramid algorithm, which is given by

$$\widetilde{V}_{j-1,t} = \sum_{l=0}^{L-1} \widetilde{h}_l \widetilde{W}_{j,t+2^{j-1}l \bmod N} + \sum_{l=0}^{L-1} \widetilde{g}_l \widetilde{V}_{j,t+2^{j-1}l \bmod N}.$$

$t = 0, 1, \dots, N-1$. The above can be expressed in matrix notation as

$$\widetilde{\mathbf{V}}_{j-1} = \widetilde{\mathbf{B}}_j^T \widetilde{\mathbf{W}}_j + \widetilde{\mathbf{A}}_j^T \widetilde{\mathbf{V}}_j.$$

If we start with $\widetilde{\mathbf{V}}_0 = \mathbf{X}$ and combine the results of the above for $j = 1, \dots, J_0$, we obtain

$$\begin{aligned} \mathbf{X} &= \widetilde{\mathbf{B}}_1^T \widetilde{\mathbf{W}}_1 + \widetilde{\mathbf{A}}_1^T \widetilde{\mathbf{B}}_2^T \widetilde{\mathbf{W}}_2 + \widetilde{\mathbf{A}}_1^T \widetilde{\mathbf{A}}_2^T \widetilde{\mathbf{B}}_3^T \widetilde{\mathbf{W}}_3 + \dots + \widetilde{\mathbf{A}}_1^T \dots \widetilde{\mathbf{A}}_{J_0-1}^T \widetilde{\mathbf{B}}_{J_0}^T \widetilde{\mathbf{W}}_{J_0} \\ &\quad + \widetilde{\mathbf{A}}_1^T \dots \widetilde{\mathbf{A}}_{J_0-1}^T \widetilde{\mathbf{A}}_{J_0}^T \widetilde{\mathbf{V}}_{J_0}. \end{aligned}$$

From the above, we can identify expressions for the MODWT details and smooth as $\widetilde{\mathcal{D}}_j = \widetilde{\mathbf{A}}_1^T \dots \widetilde{\mathbf{A}}_{j-1}^T \widetilde{\mathbf{B}}_j^T \widetilde{\mathbf{W}}_j$ and $\widetilde{\mathcal{S}}_{J_0} = \widetilde{\mathbf{A}}_1^T \dots \widetilde{\mathbf{A}}_{J_0-1}^T \widetilde{\mathbf{A}}_{J_0}^T \widetilde{\mathbf{V}}_{J_0}$. Since $\widetilde{\mathcal{D}}_j = \widetilde{\mathcal{W}}_j^T \widetilde{\mathbf{W}}_j$ and $\widetilde{\mathcal{S}}_{J_0} = \widetilde{\mathcal{V}}_{J_0}^T \widetilde{\mathbf{V}}_{J_0}$ also, we get $\widetilde{\mathcal{W}}_j = \widetilde{\mathbf{B}}_j \widetilde{\mathbf{A}}_{j-1} \dots \widetilde{\mathbf{A}}_1$ and $\widetilde{\mathcal{V}}_j = \widetilde{\mathbf{A}}_j \widetilde{\mathbf{A}}_{j-1} \dots \widetilde{\mathbf{A}}_1$.

Given the structure of $\widetilde{\mathcal{W}}_j$ and $\widetilde{\mathcal{V}}_{J_0}$, we can express the elements of $\widetilde{\mathcal{D}}_j$ and $\widetilde{\mathcal{S}}_{J_0}$ explicitly as

$$\widetilde{\mathcal{D}}_{j,t} = \sum_{l=0}^{N-1} \widetilde{h}_{j,l}^\circ \widetilde{W}_{j,t+l \bmod N} \quad \text{and} \quad \widetilde{\mathcal{S}}_{J_0,t} = \sum_{l=0}^{N-1} \widetilde{g}_{J_0,l}^\circ \widetilde{V}_{J_0,t+l \bmod N}.$$

$t = 0, 1, \dots, N-1$. By combining the above with expressions for $\widetilde{W}_{j,t}$ and $\widetilde{V}_{J_0,t}$ involving $\widetilde{h}_{j,l}^\circ$ and $\widetilde{g}_{J_0,l}^\circ$, we can argue that $\widetilde{\mathcal{D}}_j$ and $\widetilde{\mathcal{S}}_{J_0}$ can be obtained directly by filtering the time series $\mathbf{X}$ with filters whose transfer functions are given by $|\widetilde{H}_j(\cdot)|^2$ and $|\widetilde{G}_{J_0}(\cdot)|^2$. The phase functions for both these filters are identically zero, so each component of a MODWT-based MRA is associated with a zero phase filter. The zero phase property is important because it allows us to line up events in the details and smooths meaningfully with events in $\mathbf{X}$.

Finally, if the sample size for $\mathbf{X}$ is such that we can compute a level J_0 DWT, we note the following relationships between the j th level DWT and MODWT wavelet coefficients for any $j \leq J_0$:

$$W_{j,t} = 2^{j/2} \widetilde{W}_{j,2^j(t+1)-1}, \quad t = 0, \dots, N_j - 1,$$

where $N_j \equiv N/2^j$ (the above is Equation (96d)). Similarly, the scaling coefficients are related by $V_{J_0,t} = 2^{J_0/2} \widetilde{V}_{J_0,2^{J_0}(t+1)-1}$.

5.13 Exercises

- [5.1] Suppose that $\mathbf{X}$ contains a time series of even sample size N . Let $\mathcal{D}_1$ and $\mathcal{S}_1$ be the detail and smooth of an MRA of $\mathbf{X}$ based upon a $J_0 = 1$ level partial DWT. We thus have $\mathbf{X} = \mathcal{D}_1 + \mathcal{S}_1$. Let $\mathcal{D}_{\mathcal{T},1}$ and $\mathcal{S}_{\mathcal{T},1}$ be corresponding quantities for an MRA of $\mathcal{T}\mathbf{X}$ (i.e., a circular shift of $\mathbf{X}$ to the right by one unit as per Equation (52a)). We thus have $\mathcal{T}\mathbf{X} = \mathcal{D}_{\mathcal{T},1} + \mathcal{S}_{\mathcal{T},1}$ and hence $\mathbf{X} = \mathcal{T}^{-1}\mathcal{D}_{\mathcal{T},1} + \mathcal{T}^{-1}\mathcal{S}_{\mathcal{T},1}$. Let $\tilde{\mathcal{D}}_1$ and $\tilde{\mathcal{S}}_1$ be the detail and smooth of a level $J_0 = 1$ MODWT-based MRA of $\mathbf{X}$ so that we have $\mathbf{X} = \tilde{\mathcal{D}}_1 + \tilde{\mathcal{S}}_1$. Show that

$$\tilde{\mathcal{D}}_1 = \frac{1}{2} (\mathcal{D}_1 + \mathcal{T}^{-1}\mathcal{D}_{\mathcal{T},1}) \quad \text{and} \quad \tilde{\mathcal{S}}_1 = \frac{1}{2} (\mathcal{S}_1 + \mathcal{T}^{-1}\mathcal{S}_{\mathcal{T},1}).$$

Argue that we can also write

$$\tilde{\mathcal{D}}_1 = \frac{1}{N} \sum_{n=0}^{N-1} \mathcal{T}^{-n} \mathcal{D}_{\mathcal{T}^n,1} \quad \text{and} \quad \tilde{\mathcal{S}}_1 = \frac{1}{N} \sum_{n=0}^{N-1} \mathcal{T}^{-n} \mathcal{S}_{\mathcal{T}^n,1},$$

where $\mathcal{D}_{\mathcal{T}^n,1}$ and $\mathcal{S}_{\mathcal{T}^n,1}$ constitute the DWT-based MRA for $\mathcal{T}^n\mathbf{X}$. (This interpretation of the MODWT details and smooth as averages over shifted DWT details and smooth for shifted versions of $\mathbf{X}$ holds in general in that, when $N = 2^J$ and $1 \leq j \leq J_0 \leq J$, we have both

$$\tilde{\mathcal{D}}_j = \frac{1}{2^j} \sum_{n=0}^{2^j-1} \mathcal{T}^{-n} \mathcal{D}_{\mathcal{T}^n,j} \quad \text{and} \quad \tilde{\mathcal{S}}_{j_0} = \frac{1}{2^{j_0}} \sum_{n=0}^{2^{j_0}-1} \mathcal{T}^{-n} \mathcal{S}_{\mathcal{T}^n,j_0} \quad (204)$$

and also

$$\tilde{\mathcal{D}}_j = \frac{1}{N} \sum_{n=0}^{N-1} \mathcal{T}^{-n} \mathcal{D}_{\mathcal{T}^n,j} \quad \text{and} \quad \tilde{\mathcal{S}}_{j_0} = \frac{1}{N} \sum_{n=0}^{N-1} \mathcal{T}^{-n} \mathcal{S}_{\mathcal{T}^n,j_0}.$$

The idea of forming DWTs for the various shifts of a time series and then averaging together properly shifted syntheses based upon these DWTs is the key concept behind ‘cycle spinning’ as discussed in Coifman and Donoho, 1995.)

- [5.2] Show that $\tilde{\mathcal{P}}_1^T$ is the Moore–Penrose generalized inverse for $\tilde{\mathcal{P}}_1$ (Peterson, 1996). Hint: you may use the result that, if $\tilde{\mathcal{M}}$ is an $N \times 2N$ matrix such that $\tilde{\mathcal{M}}\tilde{\mathcal{P}}_1 = I_N$ and $\tilde{\mathcal{P}}_1\tilde{\mathcal{M}}$ is symmetric, then $\tilde{\mathcal{M}}$ is the Moore–Penrose inverse for $\tilde{\mathcal{P}}_1$ (Rao and Mitra, 1971).
- [5.3] Equation (176) displays $\tilde{\mathcal{B}}_2$ for the case $N = 12$ and $L = 4$. What would $\tilde{\mathcal{B}}_3$ look like for this case?
- [5.4] Using the pseudo-code in item [1] of the Comments and Extensions to Section 5.5, implement the MODWT pyramid algorithm and its inverse in your favorite computer language. Using the D(4) wavelet and scaling filters, compute and plot (or list the values for) $\tilde{\mathbf{W}}_k$ and $\tilde{\mathbf{V}}_k$, $k = 1, 2$ and 3 , for the 16 point time series $\{X_{1,t}\}$ (see caption to Figure 42). Verify your implementation of the inverse MODWT pyramid algorithm by showing that you can reconstruct $\tilde{\mathbf{V}}_{k-1}$ from $\tilde{\mathbf{W}}_k$ and $\tilde{\mathbf{V}}_k$ for $k = 1, 2$ and 3 (here we define $\tilde{\mathbf{V}}_0 = \mathbf{X}$). Verify that, by appropriately subsampling and rescaling $\tilde{\mathbf{W}}_k$ and $\tilde{\mathbf{V}}_k$, you can obtain the DWT $\mathbf{W}_k$ and $\mathbf{V}_k$ computed in Exercise [4.17].

- [5.5] For the same time series X_t considered in Exercise [4.19], compute and plot the MODWT details $\tilde{D}_k$ for $k = 1$ to 6 using any two wavelets of the following four wavelets: the Haar, D(4), C(6) and LA(8). Explain the appearance of the detail series with regard to (a) how $\{X_t\}$ is defined and (b) the particular wavelets used.
- [5.6] For the 16 point time series $\mathbf{X} = [X_{1,0}, X_{1,1}, \dots, X_{1,15}]^T$ listed in the caption to Figure 42, compute a level $J_0 = 2$ MRA based upon the Haar DWT, and compute corresponding MRAs for the circularly shifted series $\mathcal{T}^n \mathbf{X}$, $n = 1, 2$ and 3. Let $\mathcal{D}_{\mathcal{T}^n,1}$, $\mathcal{D}_{\mathcal{T}^n,2}$ and $\mathcal{S}_{\mathcal{T}^n,2}$ denote the details and smooth of the MRAs for $\mathcal{T}^n \mathbf{X}$, $n = 0, \dots, 3$. Compute also the MODWT details $\tilde{D}_1, \tilde{D}_2$ and smooth $\tilde{S}_2$ for $\mathbf{X}$. Verify numerically that Equation (204) holds.
- [5.7] Show that $\|\tilde{D}_j\|^2 \leq \|\tilde{\mathbf{W}}_j\|^2$ and that $\|\tilde{S}_j\|^2 \leq \|\tilde{\mathbf{V}}_j\|^2$ for all j .
- [5.8] Show that $\tilde{\mathcal{B}}_j^T \tilde{\mathcal{B}}_j + \tilde{\mathcal{A}}_j^T \tilde{\mathcal{A}}_j = I_N$ holds for all j (this type of result is sometimes called a ‘resolution of identity’).
- [5.9] Here we consider three complements to Exercise [171b].
- (a) Show that, if the sample size is a power of two, i.e., $N = 2^J$, and if we select $J_0 \geq J$, then all the MODWT scaling coefficients in $\tilde{\mathbf{V}}_{J_0}$ are in fact equal to $\bar{X}$, whereas, if N is not a power of two, then there can be some variation amongst the elements of $\tilde{\mathbf{V}}_{J_0}$. We can use this to argue that Equation (171a) reduces to

$$\hat{\sigma}_X^2 = \frac{1}{N} \sum_{j=1}^{J_0} \|\tilde{\mathbf{W}}_j\|^2 \text{ when } N = 2^J \text{ and } J_0 \geq J.$$

- (b) Show that, for any sample size N and any level J_0 , the sample mean of the elements of the MODWT smooth $\tilde{S}_{J_0}$ is equal to $\bar{X}$, whereas the sample mean of any MODWT detail $\tilde{D}_j$ is zero for any level j .
- (c) Show that, if $N = 2^J$ and if we select $J_0 \geq J$, then all the elements of $\tilde{S}_{J_0}$ are in fact equal to $\bar{X}$, whereas, if N is not a power of two, then there can be some variation amongst the elements of $\tilde{S}_{J_0}$.
- [5.10] Show that, for the Haar wavelet filter, we have $\tilde{H}(f) + \tilde{G}(f) = 1$. Demonstrate that this resolution of identity can be used to formulate an additive decomposition of a time series based directly on the MODWT Haar wavelet coefficients (C. A. Greenhall, private communication). What are the advantages and disadvantages of this approach as compared to the usual Haar MODWT MRA?
- [5.11] Show that, for all the LA and coiflet wavelet filters discussed in Section 4.8 and Section 4.9, respectively, we always have $L_j - 2 \geq |\nu_j^{(H)}|$ and $L_j - 2 \geq |\nu_j^{(G)}|$, where $\nu_j^{(H)}$ and $\nu_j^{(G)}$ are defined in Equations (114b) and (114a) (this exercise verifies that the limits for t given in Equations (198b) and (198c) make sense).